

# From Yang–Baxter to Robinson–Schensted–Knuth

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## Abstract

We explain how to derive the Robinson–Schensted–Knuth (RSK) correspondence, a fundamental bijection in algebraic combinatorics, from the Yang–Baxter equation. The Yang–Baxter equation arose in the study of quantum many-body systems and later became a cornerstone of the theory of solvable lattice models, particularly vertex models. In a vertex model, arrows occupy the edges of a grid, and each vertex carries a Boltzmann weight determined by the arrows on the four edges meeting at it. The weight of a configuration is the product of these local weights, and a partition function is the sum of the weights of all configurations with prescribed boundary conditions. For a vertex model whose partition functions are the Schur polynomials, the two sides of each instance of the Yang–Baxter equation admit exactly one weight-preserving matching of their summands. Carried across a grid, this forced matching is the classical RSK correspondence in the form of Fomin’s growth diagrams. In natural coordinates the local matching becomes the combinatorial three-dimensional  $R$ , a set-theoretic solution of the Zamolodchikov tetrahedron equation. Its periodic closure returns the combinatorial  $R$ -matrices of one-row crystals.

The forced matching is special to the Schur weights. For the Hall–Littlewood and  $q$ -Whittaker deformations and their spin versions, at generic parameter values no deterministic matching works for all boundary data. Reading each instance of the Yang–Baxter equation probabilistically, we replace the matching by a coupling of the two sides — a bijectivization, or probabilistic bijection — and obtain Markov operators that transport probability measures attached to vertex models. Iterated over the grid, these operators produce randomized RSK-type dynamics and interacting particle systems, including  $q$ -PushTASEP and the stochastic six-vertex model.

This survey grew out of lectures given at Institut Mittag-Leffler in the summer of 2026.

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# 1 Introduction

## 1.1 Overview

The Robinson–Schensted–Knuth (RSK) correspondence and the Yang–Baxter equation (YBE) come from different subjects: the first is a bijection of algebraic combinatorics due to Robinson [Rob38], Schensted [Sch61], and Knuth [Knu70], the second an identity of exactly solvable lattice models, see Baxter [Bax07] and Reshetikhin [Res10]. These notes explain how the first follows from the second (this derivation first appeared in Mucciconi–Petrov [MP22, §5]). A *vertex model* assigns a weight to every vertex of a grid according to the states of the four edges meeting there; the weight of a configuration is the product of these local weights, and a *partition function* — a generating function over configurations — is the sum of the weights of all configurations with prescribed boundary conditions. Figure 1 (top) shows a configuration of one such vertex model. Each row of the grid carries its own variable, called its *spectral parameter*. The *Yang–Baxter equation* of Yang [Yan67] and Baxter [Bax72] (Figure 1, bottom) equates the partition functions of two three-vertex configurations that differ by the position of a cross vertex. Between the two sides of the equation, the two rows keep their spectral parameters but meet the vertical line in the opposite order. Originally, the YBE was discovered in the study of quantum integrability, which includes the six-vertex model (exactly solved by Lieb [Lie67]), higher-spin models, the quantum inverse scattering method and algebraic Bethe ansatz of the Leningrad school (see Faddeev [Fad98]), and the quantum groups of Drinfeld [Dri87] and Jimbo [Jim85], whose  $R$ -matrices produce the vertex weights.

We consider the YBE for the vertex model whose partition functions are the *Schur polynomials* (the ordinary or, more generally, the skew ones), as in Macdonald [Mac95] and Stanley [Sta01]; see Section 2. The presentation of Schur polynomials as free fermionic lattice partition functions is classical, see Zinn-Justin [ZJ09].<sup>1</sup> For fixed boundary conditions, each of the two sides of the YBE is a sum of monomials in the two spectral parameters, and within each side these monomials are pairwise distinct (see Figure 3). Therefore the two sides admit exactly one weight-preserving matching of their summands. Performed at every cell of a square grid, this unique matching is the classical RSK correspondence in Fomin’s growth-diagram form.

Uniqueness of the matching is special to the Schur weights. For a general YBE the summands of the two sides are not matched term by term; one couples the two sides instead by a pair of Markov transitions, a construction called *bijectionization*, introduced by Bufetov–Petrov [BP19], which we take up in Section 5. The deterministic matching just described is its degenerate case, the one in which all transition probabilities are 0 or 1.

**Why rederive RSK from the Yang–Baxter equation?** The YBE is a local identity that makes a vertex model solvable. It may seem surprising that the global RSK bijection is its direct consequence. Moreover, unlike the bijection itself, the derivation from the YBE admits interesting deformations. We pay special attention to the  $q$ - and  $t$ -deformations (the  $q$ -Whittaker and Hall–Littlewood weights of Section 6): on the YBE side the deformed weights look very similar to the Schur ones, but on the RSK side, where the deterministic matching is lost, they lead to *randomized* RSK-type insertions and interacting particle systems.

We assume basic familiarity with RSK and Schur symmetric polynomials. No familiarity with vertex models or quantum integrability is needed: we define in the text every notion from that

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<sup>1</sup>Its deformation (a Schur polynomial times a deformed Weyl denominator) first appeared in Tokuyama [Tok88] as a generating function over strict Gelfand–Tsetlin patterns, with classical-group analogues in Hamel–King [HK05]. These identities were recast by Brubaker–Bump–Friedberg [BBF11] as solvable six-vertex partition functions obeying the YBE, connecting the subject to  $p$ -adic representation theory.

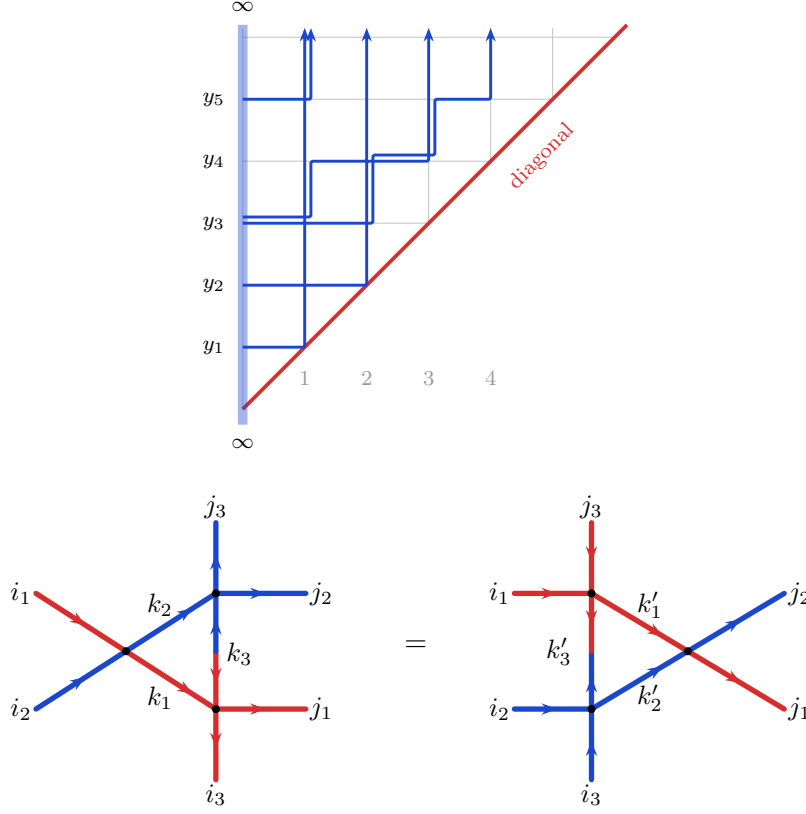


Figure 1: Top: a configuration of one of the vertex models of these notes (Section 2.4): paths enter on the left and travel up and to the right, each row of the grid carries its spectral parameter  $y_j$ , and the weight of the configuration is the product of the local vertex weights. Bottom: the Yang–Baxter equation (Section 2.5): the number of arrows along each of the six boundary edges  $i_1, i_2, i_3, j_1, j_2, j_3$  is fixed, the arrow counts  $k_1, k_2, k_3$  and  $k'_1, k'_2, k'_3$  on the internal edges are summed over, and the two partition functions are equal.

side that our arguments use, while a few standard terms appear only as references to the literature. This survey grew out of overview lectures given at the conference “Solvable lattice models, representation theory of quantum groups and algebraic combinatorics” at Institut Mittag-Leffler, Djursholm, Sweden, July 27–31, 2026.

## 1.2 A worked example

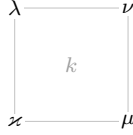
RSK is a bijection between matrices with nonnegative integer entries and pairs  $(P, Q)$  of semistandard Young tableaux of the same shape, see Knuth [Knu70], Fulton [Ful97], and Stanley [Sta01]. It reads, for example,

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix} \longleftrightarrow P = \begin{array}{|c|c|c|c|} \hline 1 & 1 & 2 & 2 \\ \hline 2 & & & \\ \hline 3 & & & \\ \hline \end{array}, \quad Q = \begin{array}{|c|c|c|c|} \hline 1 & 1 & 2 & 3 \\ \hline 2 & & & \\ \hline 3 & & & \\ \hline \end{array}.$$

Recall Fomin’s growth-diagram form of RSK [Fom88], [Fom95a]: the vertices of an  $N \times N$  grid are labeled by partitions, the labels at adjacent vertices interlace (that is, differ by a *horizontal strip*, the part of a semistandard tableau occupied by a given entry; written  $\mu \prec \lambda$ ), the left and

bottom boundaries carry the empty partitions, and each unit cell carries a nonnegative integer, the corresponding entry of the input matrix. The two output tableaux are read off the top and the right boundary of the grid. See Figure 2 for an example involving the same  $3 \times 3$  matrix as above. Expository accounts of growth diagrams include Roby [Rob91], Krattenthaler [Kra06], and Appendix 1 (by Fomin) to Chapter 7 of Stanley [Sta01].

Let us examine a single cell. It looks as follows:



The bottom-left, top-left, and bottom-right corners carry partitions which are already known from the previous cells. The entry  $k$  inside the cell is the corresponding entry of the input matrix. The top-right partition  $\nu$  is uniquely determined from this data via

$$\nu_1 = k + \max(\lambda_1, \mu_1), \quad \nu_{c+1} = \max(\lambda_{c+1}, \mu_{c+1}) + \min(\lambda_c, \mu_c) - \kappa_c \quad (c \geq 1). \quad (1.1)$$

With  $\lambda$  and  $\mu$  held fixed, (1.1) is a bijection between the pairs  $(\kappa, k)$  with  $\kappa \prec \lambda$ ,  $\kappa \prec \mu$  and  $k \in \mathbb{Z}_{\geq 0}$ , and the partitions  $\nu$  with  $\nu \succ \lambda$ ,  $\nu \succ \mu$ . Identity (1.1) is the local rule of the RSK which is sometimes referred to as the *toggle rule*, cf. Hopkins [Hop14]. We return to it in Section 3.1.

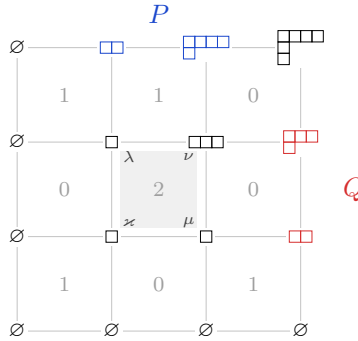


Figure 2: The growth diagram of a  $3 \times 3$  input matrix. Each vertex carries a partition, drawn as its Young diagram; the left and bottom boundaries are empty; the top and right boundary chains encode the output tableaux  $P$  and  $Q$ . In the shaded cell the corners are  $\kappa = (1)$ ,  $\lambda = (1)$ ,  $\mu = (1)$  in the local rule (1.1) with  $k = 2$ , which yields  $\nu = (3)$ .

Let us take an example. To better illustrate the matching, we choose  $\lambda$  and  $\mu$  bigger than the partitions in Figure 2: say  $\lambda = (7, 3)$  and  $\mu = (6, 4)$ , with the entry  $k = 2$  inside the cell. To keep the lists short, we restrict the bottom-left corner to a single row,  $\kappa = (\kappa_1)$ . The interlacing  $\kappa \prec \lambda$  means  $7 \geq \kappa_1 \geq 3$ , and  $\kappa \prec \mu$  means  $6 \geq \kappa_1 \geq 4$ ; together they leave exactly three admissible corners,  $\kappa_1 \in \{4, 5, 6\}$ . For each of them the rule (1.1) yields  $\nu_1 = 2 + \max(7, 6) = 9$ ,  $\nu_2 = \max(3, 4) + \min(7, 6) - \kappa_1 = 10 - \kappa_1$ , and  $\nu_3 = 0 + \min(3, 4) - 0 = 3$ ; that is,  $\nu = (9, 10 - \kappa_1, 3)$ . In the opposite direction, a top-right corner  $\nu$  with  $\nu_1 = 9$  and  $\nu_3 = 3$  interlaces with both  $\lambda$  and  $\mu$  exactly when  $4 \leq \nu_2 \leq 6$ . This yields three admissible top-right corners with  $\nu_2 \in \{4, 5, 6\}$ , as it should be by the toggle bijection  $(\kappa, k) \mapsto \nu$ .

In the simplified example, one degree of freedom remains on each side of the toggle bijection: the input corner varies through  $\kappa_1 \in \{4, 5, 6\}$ , and the output partition through  $\nu_2 \in \{4, 5, 6\}$ . As we explain in these notes, one such degree of freedom corresponds to a single instance of the YBE.

Generically, the full map  $(\varkappa, k) \mapsto \nu$  would correspond to a whole train of Yang–Baxter moves, which we explain in Section 3.2.

To write down the YBE for our example, we need spectral parameters  $x, y$  (which are exchanged by the YBE). This instance of the YBE reads

$$\underbrace{x^4 y^3 + x^5 y^4 + x^6 y^5}_{\varkappa_1=6, 5, 4} = \underbrace{x^4 y^3 + x^5 y^4 + x^6 y^5}_{\nu_2=4, 5, 6}, \quad (1.2)$$

with one summand for each admissible  $\varkappa_1$  on the left, and one for each admissible  $\nu_2$  on the right. Figure 3 draws all six configurations of this instance. The boundary occupations are the same in all six pictures and are fixed by the data of the example:  $\nu_1 - \mu_1 = 3$  and  $\nu_1 - \lambda_1 = 2$  arrows enter on the left,  $\lambda_2 - \varkappa_2 = 3$  and  $\mu_2 - \varkappa_2 = 4$  leave on the right, and the vertical line carries  $\lambda_1 - \lambda_2 = 4$  arrows at the bottom and  $\mu_1 - \mu_2 = 2$  at the top. Only the three internal occupations vary. The vertex weights behind these monomials are written out in Sections 2.4 and 2.5, see (2.6), (2.7), and (2.8); the caption of Figure 3 briefly describes them in words.

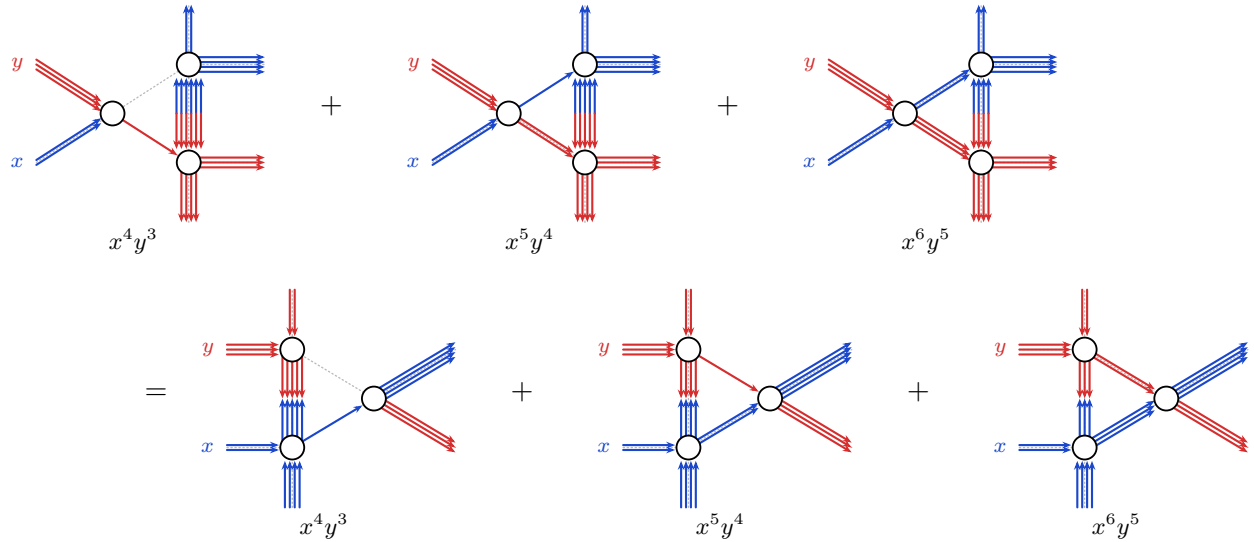


Figure 3: All six configurations of the YBE instance (1.2); below each configuration is its monomial weight. The blue line carries the spectral parameter  $x$  and the red line  $y$ . The weight of a configuration is the product of its three vertex weights: each of the two vertices on the vertical line contributes the parameter of its line ( $x$  or  $y$ ) raised to the number of arrows going out to the right, and the cross vertex contributes  $xy$  raised to the minimum of its two outgoing arrow counts.

As an identity between polynomials, (1.2) is a triviality. However, as each summand corresponds to a choice of  $\varkappa_1$  or  $\nu_2$  on the left or right, matching these summands term by term produces our toggle bijection:

$$\varkappa_1 = 6 \longleftrightarrow \nu_2 = 4, \quad \varkappa_1 = 5 \longleftrightarrow \nu_2 = 5, \quad \varkappa_1 = 4 \longleftrightarrow \nu_2 = 6.$$

This is the same matching as in (1.1), but now it is seen as a consequence of the YBE from Figure 1. The general statement of this bijection is Proposition 3.5.

### 1.3 The combinatorial 3D $R$ as a set-theoretic tetrahedron map

The matching in the example in Section 1.2 moved a single part ( $\varkappa_1$  or  $\nu_2$ ) because the fixed corners  $\lambda$  and  $\mu$  froze everything else. Let us extract the map that the matching performs in

general, when the boundary data may vary as well. Given the four occupations  $i_1, i_2, i_3, j_3$ , which are the same on the two sides (Figure 1, bottom), a summand on the left-hand side is determined by the two outgoing horizontal occupations  $j_1$  and  $j_2$  together with the smaller of the two internal ones,  $h = \min(k_1, k_2)$ . We record these as the triple  $(a, h, c) = (j_1, h, j_2)$ . On the right, a summand is determined by the two internal occupations  $k'_1$  and  $k'_2$  together with  $m = \min(j_1, j_2)$ . We record these as the triple  $(a', m, c') = (k'_1, m, k'_2)$ .

In these coordinates the unique weight-preserving matching between the two sides of the YBE reads

$$(a, h, c) \mapsto (h + \max(a - c, 0), \min(a, c), h + \max(c - a, 0)). \quad (1.3)$$

For the middle terms in the example in Figure 3, this gives  $(3, 1, 4) \mapsto (1 + 0, 3, 1 + 1) = (1, 3, 2)$ . This map is the *combinatorial three-dimensional R* of Kuniba–Okado [KO12] and Kuniba [Kun16], which solves the set-theoretic Zamolodchikov tetrahedron equation [Zam81]. We discuss this in more detail in Section 4.

Note that the map (1.3) is not a set-theoretic Yang–Baxter map in the usual sense: it acts on triples and not on pairs, and the tetrahedron equation it satisfies lives one dimension above the Yang–Baxter equation. Genuine set-theoretic Yang–Baxter maps do come out of (1.3), for instance after its third coordinate is closed into a periodic loop, which turns it into the combinatorial  $R$ -matrices of one-row crystals which also power the box-ball system of Takahashi–Satsuma [TS90]; see also the survey by Inoue–Kuniba–Takagi [IKT12]. Keeping the middle coordinate open instead, as in these notes, turns (1.3) into Fomin’s growth rule and RSK. Thus, the box-ball system and RSK are two global readings of one local map (Section 4.3). For a direct combinatorial link between the box-ball system and RSK, see Fukuda [Fuk04].

## 1.4 From Yang–Baxter to interacting particle systems

The second half of these notes deforms the constructions we outlined in Section 1.2. At the level of symmetric functions, the Schur polynomials are replaced either by the Hall–Littlewood polynomials or by the  $q$ -Whittaker polynomials, which are two different one-parameter specializations of the Macdonald polynomials [Mac95]. In the Yang–Baxter equations for the corresponding deformed weights, the two sides no longer admit a term-by-term bijective matching. Instead, one can produce a *randomized* matching (a *bijectivization* of the YBE), which is a conditional distribution (or a Markov transition kernel) assigning a random term from the right-hand side to each term of the left-hand side (or vice versa, a random term from the left-hand side to each term of the right-hand side).

For the instance of the YBE from Figure 3 with the  $q$ -Whittaker deformed weights, at  $q = \frac{1}{2}$ , the six summands become proportional to

$$\underbrace{651}_{\kappa_1=6} + \underbrace{(217 + 868xy)}_{\kappa_1=5} + \underbrace{(4 + 168xy + 448(xy)^2)}_{\kappa_1=4} = \underbrace{620}_{\nu_2=4} + \underbrace{(245 + 840xy)}_{\nu_2=5} + \underbrace{(7 + 196xy + 448(xy)^2)}_{\nu_2=6}.$$

The deformed summands for general  $q$  are written out in Section 7. The two sides are equal as polynomials in  $xy$ , but no summand on the left equals a summand on the right. The independent coupling, which chooses a term on the right with probability proportional to its weight, sends  $\kappa_1 = 5$  to  $\nu_2 = 4, 5, 6$  with probabilities proportional to the three summands on the right. At  $x = y = \frac{1}{2}$ , these probabilities are  $\frac{620}{1159}, \frac{455}{1159}, \frac{84}{1159}$ , respectively.

The resulting randomized RSKs obtained from bijectivizations of the YBE are related to known integrable interacting particle systems. Let us survey the literature on these constructions. The totally asymmetric simple exclusion process (TASEP), the simplest example of an integrable interacting particle system, was introduced in biology by MacDonald–Gibbs–Pipkin [MGP68] and in probability by Spitzer [Spi70] (see also Liggett [Lig75] for the general theory of interacting particle systems, and Rost [Ros81] for the limit shape of TASEP started from the step initial configuration). Schur functions and RSK found applications to particle systems around 2000 starting from the work of Johansson [Joh00], though they entered probability much earlier: Vershik–Kerov [VK77] and Logan–Shepp [LS77] found the limit shape of the Plancherel random partition already in 1977.

Schur measures and processes, introduced by Okounkov [Oko01] and Okounkov–Reshetikhin [OR03], are probability distributions on partitions (or sequences of partitions) whose probability weights are expressed through Schur polynomials. Through RSK, Schur measures are connected to TASEP, last-passage percolation, and longest increasing subsequences of random permutations, see Baik–Deift–Johansson [BDJ99], Borodin–Okounkov–Olshanski [BOO00], and Johansson [Joh00]. Borodin and Ferrari [BF14] constructed two-dimensional growth dynamics on Schur processes. These dynamics contain TASEP as a one-dimensional marginal.

Following Borodin–Ferrari, other symmetric functions from the Macdonald hierarchy [Mac95] give rise to deformed particle systems such as the  $q$ -TASEP pioneered by Borodin–Corwin [BC14]. Randomized RSK-type dynamics for these deformations predate Yang–Baxter bijectivizations: such dynamics were constructed by O’Connell–Pei [OP13] and Borodin–Petrov [BP16], and then by Bufetov–Petrov [BP15] in the Hall–Littlewood case and Matveev–Petrov [MP17] in the full  $q$ -Whittaker case (including discrete variants of  $q$ -TASEP and  $q$ -PushTASEP), all without using the YBE.

Stochastic vertex models entered probability in Borodin [Bor17], Corwin–Petrov [CP16], and Borodin–Petrov [BP18]. Bijectivization, first used in Bufetov–Petrov [BP19], turns the YBE itself into local stochastic moves. It was developed further in [BMP21], [MP22], [PS22], [PS24], [NP23]. The Yang–Baxter bijectivization serves as a bridge between vertex models and interacting particle systems: directly, through scalar marginals, or through degenerations, it connects to  $q$ -TASEP and  $q$ -PushTASEP, to the stochastic six-vertex model and ASEP [BCG16], [BBW16], [BM18], and to directed random polymers [Sep12], [OY01], [BC16]. The symmetric functions play a secondary role, and the YBE side can even produce new families, such as the spin  $q$ -Whittaker polynomials [BW21], [MP22], [Kor24]. See [Cor14] and [BGW22] for charts of hierarchies of integrable models. In these notes, Section 5 presents the general bijectivization, and Section 7 sketches some of the resulting particle systems.

Let us also mention that in rigorous two-dimensional statistical mechanics, the random star-triangle transformation (itself a bijectivization of the YBE) underlies the proof of rotational invariance of the critical Ising, FK, and six-vertex models in Duminil-Copin–Kozłowski–Krachun–Manolescu–Oulamara [DCKK<sup>+</sup>20].

## 1.5 Outline

The notes are organized as follows. Section 2 presents the three vertex models whose partition functions are the Schur polynomials, together with the YBE and the Cauchy identity. In Section 3, we match the two sides of the YBE term by term and identify this matching with RSK in Fomin’s growth-diagram form. Section 4 recasts the same matching in the coordinates of (1.3) as the combinatorial three-dimensional  $R$  satisfying the tetrahedron equation, and discusses the periodic closure and the one-row crystals powering the box-ball system. In Section 5, we describe the general

bijectivization procedure, of which the matching of Section 3 is the deterministic case. Section 6 treats the  $t$ - and  $q$ -deformations of the Schur vertex models (Hall–Littlewood and  $q$ -Whittaker), related by Macdonald duality and fusion. Section 7 sketches how the deformed weights, combined with bijectivization, yield integrable stochastic interacting particle systems. Section 8 discusses further directions.

A reader who wants only the derivation of RSK from the Yang–Baxter equation may read the Introduction and then Sections 2 and 3, and stop. Section 4 adds the three-dimensional point of view connected to the set-theoretic tetrahedron equation. The sections after it form the probabilistic half of the notes powered by  $q$ - and  $t$ - deformations, where the matching of the two sides of the YBE is no longer deterministic.

## 1.6 Acknowledgements

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## 2 Vertex models for Schur polynomials

### 2.1 Conventions

A *signature* of length  $N$  is a weakly decreasing integer vector  $\lambda = (\lambda_1 \geq \dots \geq \lambda_N)$ ,  $\lambda_i \in \mathbb{Z}$ ; the set of these is  $\text{Sign}_N$ . A *partition* is a signature with  $\lambda_N \geq 0$ , identified with its Young diagram (the left-justified array of  $\lambda_i$  boxes in row  $i$ , drawn top to bottom in the English convention);  $|\lambda| = \sum_i \lambda_i$  is its size,  $\ell(\lambda)$  its number of nonzero parts, and  $s_\lambda(x_1, \dots, x_n)$  is the Schur polynomial. We write  $\mathbb{Z}_{\geq 0} = \{0, 1, 2, \dots\}$ . The *conjugate* (or transpose) partition  $\lambda'$  is the reflection of the diagram across the main diagonal,

$$\lambda'_i = \#\{j : \lambda_j \geq i\} \quad (\text{the height of the } i\text{th column of } \lambda),$$

so that  $(\lambda')' = \lambda$  and  $|\lambda'| = |\lambda|$ . Our running example is

$$\lambda = (4, 4, 1), \quad \lambda' = (3, 2, 2, 2),$$

displayed in Figure 4.

*Strips and interlacing.* Two partitions  $\mu \subseteq \lambda$  (diagram inclusion) differ by a *horizontal strip* if their skew shape  $\lambda/\mu$  has at most one box in each *column*, and by a *vertical strip* if  $\lambda/\mu$  has at most one box in each *row*. These two conditions are exchanged by conjugation, and each is encoded

by an interlacing relation on signatures. We write  $\mu \prec \lambda$  (equivalently  $\lambda \succ \mu$ ) for the *horizontal* interlacing

$$\lambda_1 \geq \mu_1 \geq \lambda_2 \geq \mu_2 \geq \cdots,$$

which for partitions says exactly that  $\lambda/\mu$  is a horizontal strip. The *transposed* (vertical) interlacing is its conjugate,

$$\mu \prec' \lambda \stackrel{\text{def}}{\iff} \mu' \prec \lambda', \quad \text{i.e.} \quad \lambda'_1 \geq \mu'_1 \geq \lambda'_2 \geq \mu'_2 \geq \cdots.$$

See Figure 5 for an illustration.

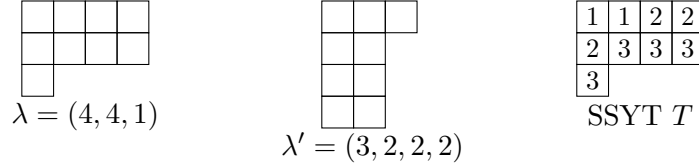


Figure 4: The partition  $\lambda = (4, 4, 1)$  (left), its conjugate  $\lambda' = (3, 2, 2, 2)$  obtained by reflecting across the main diagonal (middle), and a semistandard Young tableau  $T$  of shape  $\lambda$  with entries in  $\{1, \dots, n\}$ ,  $n = 3$  (right): rows weakly increase left to right, columns strictly increase top to bottom. The cells filled with a given entry  $k$  form a horizontal strip (at most one per column), so  $T$  records the horizontal-strip interlacing sequence  $\emptyset \prec (2) \prec (4, 1) \prec (4, 4, 1)$ . The weight of  $T$  is  $x^{\text{wt}(T)} = x_1^2 x_2^3 x_3^4$ , and  $s_\lambda = \sum_T x^{\text{wt}(T)}$  is the sum of weights over all semistandard tableaux.



Figure 5: Left: a horizontal strip  $\mu \prec \lambda$  (shaded), at most one box per column. Right: a vertical strip  $\mu \prec' \lambda$  (shaded), at most one box per row. Conjugation exchanges the two.

*Weights.* Vertex weights are taken *nonnegative* — and that is all the Yang–Baxter bijectivization of Section 5 will require of them. A model is moreover called *stochastic* when its weights, read as conditional probabilities of the outgoing edges given the incoming ones, sum to one, so that the row-to-row transfer matrix is a Markov operator; this is a convenient extra structure used in connection with interacting particle systems. Both are claims about real parameters: whenever a statement below is probabilistic, we work in the regime  $0 \leq q < 1$ ,  $0 \leq t < 1$ , and  $x_i, y_j \geq 0$  with  $x_i y_j < 1$ . The identities themselves need no such restriction, holding as identities of rational functions or of formal series.

## 2.2 Schur vertex model I: Horizontal strips

A solvable lattice model assigns Boltzmann weights to the local configurations on the square grid, with paths (arrows) conserved at each vertex, and the weight of a whole configuration is the product of its vertex weights. The most classical instance is the *free fermion five-vertex model*, whose partition function is a Schur polynomial. This identity is classical [ZJ09]: in the equivalent phase model language (cf. Remark 2.4 below) it appears in [Tsi06], and as a six-vertex partition function governed by the YBE it was developed in [BBF11]. We describe the model concretely.

Fix  $n$  and work on the grid  $\mathbb{Z}_{\geq 1} \times \{1, \dots, n\}$ :  $n$  horizontal rows indexed from the bottom by  $i = 1, \dots, n$ , and columns indexed by a half-line coordinate. Each edge carries an occupation in  $\{0, 1\}$ , and we encode a local vertex by its four edge occupations

$$(i_1, j_1; i_2, j_2) = (\text{bottom in, left in; top out, right out}), \quad i_1, j_1, i_2, j_2 \in \{0, 1\}. \quad (2.1)$$

Path conservation forces the weight to vanish unless  $i_1 + j_1 = i_2 + j_2$ , leaving six admissible vertices; the *five-vertex* condition discards the doubly-occupied vertex  $(1, 1; 1, 1)$ . The remaining five vertices, and the Boltzmann weights attached to a vertex sitting in row  $i$ , are shown in Figure 6.

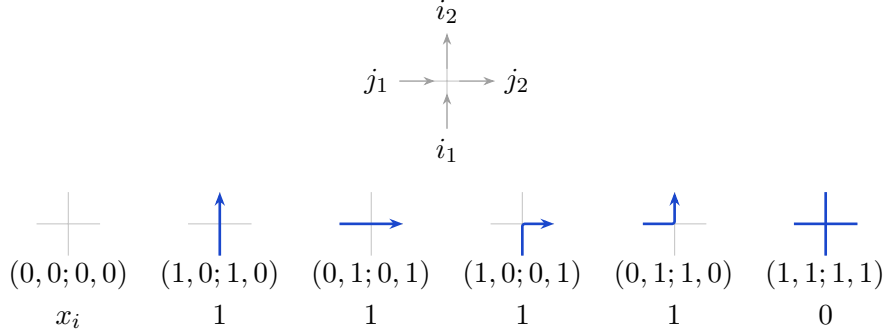


Figure 6: Top: the edge-occupation convention of (2.1) — a vertex in row  $i$  with vertical-in  $i_1$  (bottom), horizontal-in  $j_1$  (left), vertical-out  $i_2$  (top), horizontal-out  $j_2$  (right); paths run up and to the right. Bottom: the five admissible vertices with their Boltzmann weights in row  $i$ .

**Remark 2.1** (Free fermion condition). The six Boltzmann weights (the five admissible ones together with the discarded doubly-occupied vertex of weight 0) satisfy the *free fermion condition*

$$w\left(\begin{array}{c} | \\ | \\ | \\ | \end{array}\right) \cdot w\left(\begin{array}{c} | \\ | \\ | \\ | \end{array}\right) + w\left(\begin{array}{c} | \\ | \\ | \\ | \end{array}\right) \cdot w\left(\begin{array}{c} | \\ | \\ | \\ | \end{array}\right) = w\left(\begin{array}{c} | \\ | \\ | \\ | \end{array}\right) \cdot w\left(\begin{array}{c} | \\ | \\ | \\ | \end{array}\right). \quad (2.2)$$

A partition  $\lambda = (\lambda_1 \geq \dots \geq \lambda_N \geq 0)$ , padded by zeros to a fixed length  $N$ , enters as boundary data through the “parts + staircase” point set

$$\mathcal{S}(\lambda) = \{\lambda_j + N - j + 1\}_{1 \leq j \leq N} \subset \mathbb{Z}_{\geq 1}, \quad (2.3)$$

a set of  $N$  distinct occupied sites, the listed values strictly decreasing in  $j$ . A signature with a negative part would leave the half-line  $\mathbb{Z}_{\geq 1}$  on which the columns are indexed, and is excluded here. For the partition function  $Z_\lambda$  we take a single block of  $n$  rows on which  $\lambda$  has  $N = n$  parts, feed in  $n$  paths from the bottom at the positions  $\mathcal{S}(\lambda)$ , require them to exit through the right boundary at infinity (each path eventually travels horizontally, so all but finitely many vertices are weight-1 pass-throughs), and keep the top and left boundaries empty. Summing the product of weights from Figure 6 over all admissible five-vertex configurations with this boundary, by definition, is the partition function  $Z_\lambda(x_1, \dots, x_n)$ .

**Proposition 2.2** (Schur polynomial as a five-vertex partition function). *We have*

$$s_\lambda(x_1, \dots, x_n) = Z_\lambda(x_1, \dots, x_n).$$

*Sketch of proof.* One can use a straightforward bijection between five-vertex configurations and semistandard Young tableaux which preserves the weights  $x^{\text{wt}(T)}$  up to reversing the variables  $x_i \mapsto x_{n+1-i}$  (the reversal does not affect the partition function, which is symmetric in the  $x_i$ 's). See Figure 7 for an illustration of a vertex model configuration corresponding to the semistandard Young tableau from Figure 4. Note that each row of the vertex model corresponds to a horizontal strip in the tableau.  $\square$

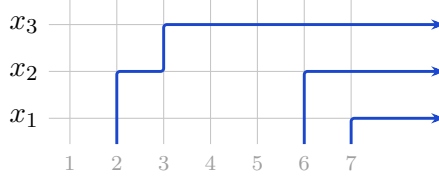


Figure 7: The five-vertex configuration corresponding to the semistandard tableau  $T$  of Figure 4 of shape  $\lambda = (4, 4, 1)$  with  $n = 3$ .

**Remark 2.3.** Fix an additional partition  $\mu$  with  $N$  parts and take  $\lambda$  with  $N + n$  parts. Reading the paths in a block of  $n$  rows between the bottom boundary  $\mathcal{S}(\lambda)$  and the top boundary  $\mathcal{S}(\mu)$ , with the left boundary empty and one path exiting through the right of each row, yields the skew Schur polynomial  $s_{\lambda/\mu}$ . The length relation is forced by the boundary flux: each of the  $n$  rows releases exactly one path to the right, so the bottom signature carries  $n$  coordinates more than the top one.

**Remark 2.4** (Higher spin version). The free fermion five-vertex model described above can be bijectively mapped to a free fermion instance of the higher spin six-vertex model studied in [Bor17], [BP18]. The allowed configurations in this model have  $i_1, i_2 \in \mathbb{Z}_{\geq 0}$  and  $j_1, j_2 \in \{0, 1\}$ . The vertex weights (in the Schur case) in the  $i$ -th row are given by

$$w(i_1, j_1; i_2, j_2) = \mathbf{1}_{i_1+j_1=i_2+j_2} x_i^{j_2}.$$

The configuration which bijectively corresponds to the one in Figure 7 is shown in Figure 8. Now the top boundary is given by the multiset  $\{\lambda_1, \lambda_2, \dots, \lambda_n\}$ , and the paths enter everywhere from the left boundary, while the bottom and the right boundaries are empty. In contrast with the five-vertex model, the columns are now indexed by  $\mathbb{Z}_{\geq 0}$  (zero parts of  $\lambda$  occupy column 0); this indexing is needed for the weights  $x_i^{j_2}$  to produce the correct total power of  $x_i$  in each row.

**Remark 2.5** (The free fermion six-vertex model). Allowing the vertex  $(1, 1; 1, 1)$  turns the five-vertex model into the *six-vertex model*. Keeping the free fermion condition (2.2), one can define free fermion Schur functions [ABPW23], [Nap24], which generalize both horizontal- and vertical-strip Young tableaux (the vertical strip setting is detailed in Section 2.4 below). This family of functions includes as particular cases the factorial and shifted Schur polynomials, as well as the Frobenius–Schur functions.

### 2.3 Skew Young diagrams and vertex model rows

Drop the condition  $j_1, j_2 \leq 1$  on the horizontal edges of the higher-spin model (Remark 2.4), keeping  $i_1, i_2 \in \mathbb{Z}_{\geq 0}$  and the arrow conservation  $i_1 + j_1 = i_2 + j_2$  at each vertex. Consider a single row of vertices on this more general vertex model. We read configurations of vertical arrows as multisets,

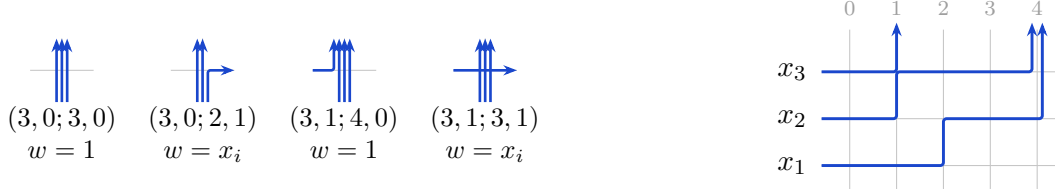


Figure 8: Left: the four vertex types  $(j_1, j_2) \in \{0, 1\}^2$ , labelled  $(i_1, j_1; i_2, j_2)$ , with weights  $w = x_i^{j_2}$ . The vertical occupation  $i_1, i_2 \geq 0$  is arbitrary, subject to  $i_1 + j_1 = i_2 + j_2$ ; here we draw the representatives with  $i_1 = 3$ . Right: the higher-spin six-vertex configuration corresponding to the tableau  $T$  (equivalently, to the five-vertex configuration of Figure 7); the columns are indexed from 0.

and interpret them as partitions  $\lambda$  (at the top) and  $\mu$  (at the bottom). In particular, the vertical occupation  $i_2$  at the top of column  $x$  is the multiplicity  $m_x(\lambda) = \#\{i : \lambda_i = x\}$ . Writing  $h_x$  for the occupation of the horizontal edge entering column  $x$  from the left, we have

$$h_x = \sum_{y \geq x} (m_y(\lambda) - m_y(\mu)) = \lambda'_x - \mu'_x.$$

The horizontal edges are thus nonnegative if and only if  $\lambda'_x \geq \mu'_x$  for all  $x$ , which is equivalent to the Young diagram inclusion  $\mu \subseteq \lambda$ . Figure 9 illustrates this dictionary.

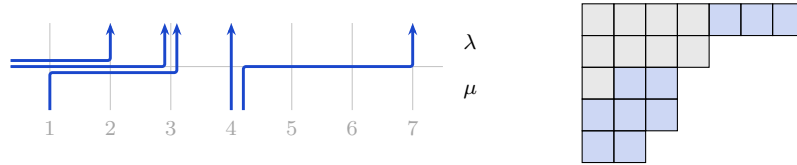


Figure 9: The dictionary between general vertex models and skew Young diagrams.

Restricting  $\lambda/\mu$  to horizontal strips recovers the higher-spin model of Remark 2.4. Restricting to vertical strips, we obtain another combinatorial condition on the allowed vertex types:

**Lemma 2.6.** *For a vertex row configuration corresponding to a skew Young diagram  $\lambda/\mu$ , the inequality  $j_2 \leq i_1$  holds at every vertex if and only if  $\lambda/\mu$  is a vertical strip.*

*Proof.* At column  $x$  the bottom vertical edge carries  $i_1 = m_x(\mu) = \mu'_x - \mu'_{x+1}$  and the outgoing horizontal edge carries  $j_2 = h_{x+1} = \lambda'_{x+1} - \mu'_{x+1}$ , so

$$j_2 \leq i_1 \iff \lambda'_{x+1} - \mu'_{x+1} \leq \mu'_x - \mu'_{x+1} \iff \lambda'_{x+1} \leq \mu'_x.$$

This completes the proof.  $\square$

The vertical-strip model is discussed in detail in Section 2.4 below.

**Remark 2.7.** The five-vertex model is not a direct specialization of the model in Remark 2.4, but it is a bijective change of coordinates from the horizontal-strip case, which corresponds to the shifted coordinates  $\mathcal{S}(\lambda)$  (2.3).

The rest of this subsection is a digression, not used later in the survey. In the shifted coordinates (2.3), the six-vertex model of Remark 2.5 (that is, with the doubly occupied vertex  $(1, 1; 1, 1)$  allowed) captures one more distinguished class of skew Young diagrams — *ribbons* (border strips): connected skew diagrams containing no  $2 \times 2$  block of boxes.

**Lemma 2.8** (Ribbon condition). *Let  $\lambda, \mu$  be partitions with at most  $N$  parts, both encoded as in (2.3) with this common  $N$ . Consider a single row of the six-vertex model (all edges carry occupations in  $\{0, 1\}$ , and all six vertices of Figure 6, including  $(1, 1; 1, 1)$ , are admissible) with bottom boundary  $\mathcal{S}(\mu)$ , top boundary  $\mathcal{S}(\lambda)$ , and empty left and right horizontal edges. An admissible configuration exists if and only if  $\mu \subseteq \lambda$  and every connected component of  $\lambda/\mu$  is a ribbon (equivalently,  $\lambda/\mu$  contains no  $2 \times 2$  block of boxes). The configuration is then unique; see Figure 10.*

*Proof.* The configuration is forced by the boundary data: the occupation  $h_x$  of the horizontal edge between columns  $x$  and  $x + 1$  must equal

$$h_x = \#(\mathcal{S}(\mu) \cap \{1, \dots, x\}) - \#(\mathcal{S}(\lambda) \cap \{1, \dots, x\}) = \#\{i: \lambda_i - i \geq x - N\} - \#\{i: \mu_i - i \geq x - N\}, \quad (2.4)$$

and any assignment with all  $h_x \in \{0, 1\}$  is admissible. Row  $i$  of  $\lambda/\mu$  contains a box of content  $x - N$  (content = column - row) if and only if  $\mu_i - i < x - N \leq \lambda_i - i$ . Hence  $h_x \geq 0$  for all  $x$  is equivalent to  $\mu \subseteq \lambda$ , and then  $h_x$  counts the boxes of  $\lambda/\mu$  of content  $x - N$ . The condition  $h_x \leq 1$  states that every diagonal of  $\lambda/\mu$  carries at most one box, which is equivalent to the absence of a  $2 \times 2$  block: two boxes of equal content force boxes of that content in all intermediate rows, and two of them in adjacent rows produce a  $2 \times 2$  block.  $\square$

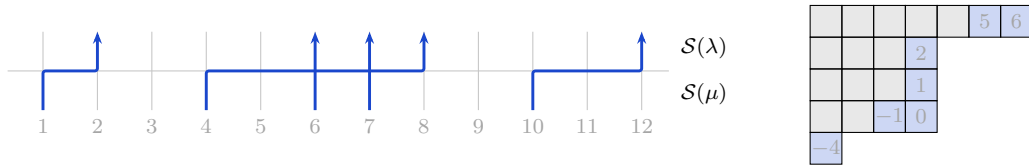


Figure 10: Left: the unique single-row six-vertex configuration with bottom  $\mathcal{S}(\mu) = \{1, 4, 6, 7, 10\}$  and top  $\mathcal{S}(\lambda) = \{2, 6, 7, 8, 12\}$ , where  $\mu = (5, 3, 3, 2)$ ,  $\lambda = (7, 4, 4, 4, 1)$ , and  $N = 5$ . Right: the disconnected skew shape  $\lambda/\mu$ , a disjoint union of three ribbons of heights 1, 3, 1, with the content of each box indicated; the components do not even share corners, being separated by the empty diagonals of contents  $-3, -2$  and  $3, 4$ . The horizontal edge between columns  $x$  and  $x + 1$  is occupied precisely when  $\lambda/\mu$  has a box of content  $x - N$ . The path emitted at column 4 jumps over the two stationary particles at columns 6 and 7 — the doubly occupied vertices  $(1, 1; 1, 1)$  — which match the two boxes with a bottom neighbor (contents 1 and 2) in the height-3 ribbon, in agreement with Remark 2.9.

Compare (2.4) with the identity  $h_x = \lambda'_x - \mu'_x$  from the beginning of this subsection: in the multiplicity coordinates the horizontal edges count the columns of  $\lambda/\mu$ , while in the shifted coordinates they count the diagonals.

**Remark 2.9** (Murnaghan–Nakayama structure). In the configuration of Lemma 2.8, the number of vertices of type  $(1, 0; 0, 1)$  equals the number of ribbon components of  $\lambda/\mu$ , and the number of doubly occupied vertices  $(1, 1; 1, 1)$  equals  $\sum_r (\text{ht}(r) - 1)$ , where  $\text{ht}(r)$  is the number of rows of the ribbon  $r$  (the doubly occupied columns correspond to the boxes of  $\lambda/\mu$  having a bottom neighbor). Thus, assigning weight  $-1$  to the vertex  $(1, 1; 1, 1)$  produces the signs of the Murnaghan–Nakayama rule [Mac95, Ch. I, §7]. Matrix elements of the free fermion six-vertex transfer matrices, with weights factorized over the ribbons of  $\lambda/\mu$ , are computed in [Nap24, §4.2]; for the Yang–Baxter algebra of the general asymmetric six-vertex model in terms of ribbons, leading to cylindric Murnaghan–Nakayama rules for Hecke characters, see [Kor21, §4.3].

## 2.4 Schur vertex model II: Vertical strips

We now describe the third Schur vertex model (labeled II in the subsection titles, since the higher-spin realization of Remark 2.4 counts as the second), which operates with the transposed Young diagram, and corresponds to vertical-strip interlacing sequences as described in Section 2.3 above. This model is the  $q = 0$  degeneration of the  $q$ -Whittaker vertex model introduced in [Kor13, §6.1], see Section 6.2 below. We follow [MP22]. This vertex model may be viewed as a “dual” to the free fermion five-vertex model of Section 2.2 (and its higher-spin equivalent version from Remark 2.4).

We encode top and bottom boundary conditions of this new vertex model by Young diagrams. Namely, to a Young diagram  $\lambda$  we associate the multiset of its column lengths via

$$\lambda_i - \lambda_{i+1} = \#\{\text{paths at the } i\text{-th vertical edge}\}, \quad i = 1, 2, \dots \quad (2.5)$$

For example, the Young diagram  $\lambda = (4, 4, 1)$  corresponds to the configuration with three vertical edges in column 2, one vertical edge in column 3, and no vertical edges in the other columns.

**Remark 2.10.** Note that the convention (2.5) used for the vertical-strip model differs from the one of Section 2.3 by conjugation. Therefore, in the vertical-strip model, we still get *horizontal-strip differences* between consecutive rows of the vertex model. In the Schur case this change of convention is immaterial: in either dictionary the partition function is a skew Schur polynomial, and only the indexing shape is transposed. However, we follow the existing literature [BW21], [MP22] which deals with deformations of the models where this conjugation is meaningful. We discuss some of the deformations in Sections 6 and 7 below.

We now define two vertex models, exchanged by the reflection  $i_1 \leftrightarrow i_2$  of the vertical axis. The *up-right* (blue) weight is

$$W(i_1, j_1; i_2, j_2) = \mathbf{1}_{i_1+j_1=i_2+j_2} \mathbf{1}_{i_1 \geq j_2} x^{j_2}, \quad (2.6)$$

with  $x$  the spectral parameter of the row, and the *down-right* (red) weight is its reflection,

$$W^*(i_1, j_1; i_2, j_2) = \mathbf{1}_{i_2+j_1=i_1+j_2} \mathbf{1}_{i_2 \geq j_2} x^{j_2}. \quad (2.7)$$

Here  $i_1, i_2, j_1, j_2 \in \mathbb{Z}_{\geq 0}$ .

Encode the bottom and top boundaries of a single row by the column-length coordinates (2.5) of two partitions  $\mu$  and  $\lambda$ . The condition  $i_1 \geq j_2$  in (2.6) (respectively  $i_2 \geq j_2$  in (2.7)) is equivalent to the requirement that  $\lambda/\mu$  be a *horizontal* strip, see Lemma 2.6 and Remark 2.10. The skew functions  $G_{\lambda/\mu}(x)$  and  $G_{\lambda/\mu}^*(x)$  are defined as the weights of the single rows with these boundary conditions (for a single row, there is only one admissible configuration in both vertex models).

Gluing  $k$  rows with spectral parameters  $y_1, \dots, y_k$  produces  $G_{\lambda/\mu}(y_1, \dots, y_k)$  and  $G_{\lambda/\mu}^*(y_1, \dots, y_k)$  in the similar manner.

**Remark 2.11.** The horizontal arrows on the left boundary in the models for  $G_{\lambda/\mu}$  and  $G_{\lambda/\mu}^*$  may be arbitrary. This may be modeled by placing an infinite number of vertical edges at column 0 with occupation  $i_1 = i_2 = \infty$ ; formally, one sets  $i_1 = M$ , lets the outgoing occupation be determined by arrow conservation, and sends  $M \rightarrow \infty$ . The two conservation laws differ: for  $W$  one has  $i_2 = M + j_1 - j_2$ , while for  $W^*$ , whose law is  $i_2 + j_1 = i_1 + j_2$ , one has  $i_2 = M + j_2 - j_1$ . Then, the weights become  $W(\infty, j_1; \infty, j_2) = x^{j_2}$  and  $W^*(\infty, j_1; \infty, j_2) = x^{j_2}$ , which are independent of  $j_1$ ; for (2.6) this value is already attained at every finite  $M \geq j_2$ , and for (2.7) at every finite  $M \geq j_1$ . These boundary weights are the correct choice for our partition functions. This convention is also equivalent to setting  $\lambda_0 = +\infty$ , see (2.5).

Configurations contributing a nonzero weight to  $G_{\lambda/\mu}$  or  $G_{\lambda/\mu}^*$  are in bijection with interlacing chains of partitions  $\mu = \lambda^{(0)} \prec \lambda^{(1)} \prec \dots \prec \lambda^{(k)} = \lambda$ , with the  $i$ -th row creating the horizontal strip  $\lambda^{(i)}/\lambda^{(i-1)}$ . Unlike for the models in Section 2.2, both  $G$  and  $G^*$  are defined for any number of variables and are stable under  $y_k \rightarrow 0$ . On the other hand, the number of variables  $k$  must be at least the length of the longest column of  $\lambda/\mu$  for the weight to be nonzero.

**Proposition 2.12.** *For partitions  $\mu \subseteq \lambda$  and any number of variables, we have*

$$G_{\lambda/\mu}(y_1, \dots, y_k) = G_{\lambda/\mu}^*(y_1, \dots, y_k) = s_{\lambda/\mu}(y_1, \dots, y_k), \quad G_\lambda := G_{\lambda/\emptyset}.$$

*Proof.* Follows by identifying vertex model configurations with chains of interlacing partitions (differing by horizontal strips), equivalently, with semistandard Young tableaux of shape  $\lambda/\mu$ . An illustration of this identification is given in Figure 11. The vertex weights (2.6) and (2.7) produce the correct weight  $y^{\text{wt}(T)}$  for each tableau  $T$  entering the (skew) Schur polynomial.  $\square$

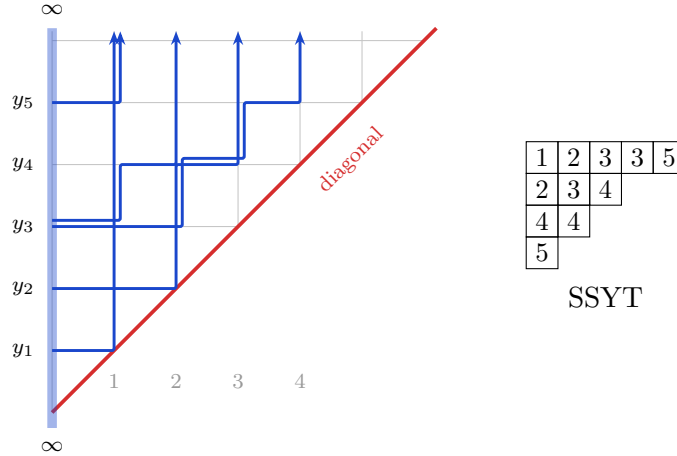


Figure 11: A configuration of the vertex model  $W$  (2.6) and the corresponding semistandard Young tableau. The shape is  $\lambda = (5, 3, 2, 1, 0)$ , and the weight is  $y_1 y_2^2 y_3^3 y_4^3 y_5^2 = y^{\text{wt}(T)}$ . No path in the vertex model can go below the diagonal thanks to the condition  $i_1 \geq j_2$  in (2.6). The thick edge at column 0 carries the infinite vertical occupation (see Remark 2.11), from which the paths peel off to the right.

## 2.5 Yang–Baxter equation and Cauchy identity

Let us now formulate the *Yang-Baxter equation* connecting the models  $W$  and  $W^*$ . Define the cross-vertex weights by

$$\mathbb{R}_{x,y}(i_1, j_1; i_2, j_2) = \mathbb{R}_{x,y} \left( \begin{array}{cc} j_1 & i_2 \\ i_1 & j_2 \end{array} \right) = \mathbf{1}_{i_1+j_2=j_1+i_2} (xy)^{\min(i_2, j_2)}, \quad (2.8)$$

with  $(i_1, j_1)$  the incoming and  $(i_2, j_2)$  the outgoing arrow counts.

**Proposition 2.13** (Yang–Baxter equation). *For all boundary occupations  $i_1, i_2, i_3, j_1, j_2, j_3 \in \mathbb{Z}_{\geq 0}$  and spectral parameters  $x, y$ , we have*

$$\sum_{k_1, k_2, k_3 \geq 0} \mathbb{R}_{x,y}(i_2, i_1; k_2, k_1) W(k_3, k_2; j_3, j_2) W^*(i_3, k_1; k_3, j_1) \\ = \sum_{k'_1, k'_2, k'_3 \geq 0} W(i_3, i_2; k'_3, k'_2) W^*(k'_3, i_1; j_3, k'_1) \mathbb{R}_{x,y}(k'_2, k'_1; j_2, j_1), \quad (2.9)$$

where  $W = W_x$  (2.6) and  $W^* = W_y^*$  (2.7) carry the row parameters  $x$  and  $y$ . Graphically, the two local vertex configurations of Figure 12 have equal partition functions: the  $\mathbb{R}$ -cross may be dragged across the  $W/W^*$  pair while preserving the six boundary edges.

*Proof.* Follows by a direct computation with explicit weights. A term-by-term verification, with the summands and their admissible ranges written out, is given in Lemma 3.2 below.  $\square$

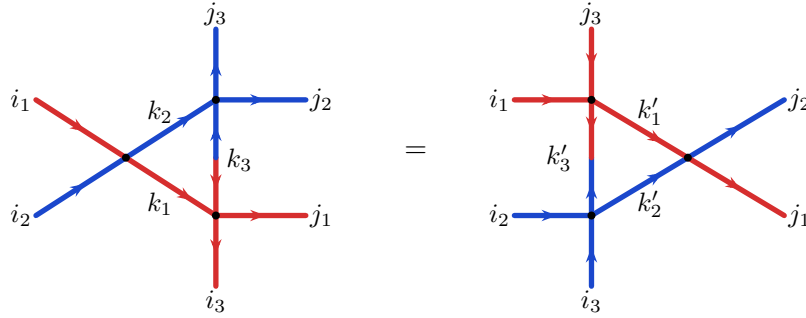


Figure 12: The Yang–Baxter equation (2.9) of Proposition 2.13. The boundary arrow counts  $i_1, i_2, i_3, j_1, j_2, j_3 \in \mathbb{Z}_{\geq 0}$  are fixed, and the internal arrow counts  $k_1, k_2, k_3$  and  $k'_1, k'_2, k'_3$  are summed over. The vertical edge on the left or on the right serves as a source or a sink, respectively.

Let us now use the weights  $W$  and  $W^*$  to construct a vertex model for the sum

$$\sum_{\lambda} s_{\lambda}(x_1, \dots, x_N) s_{\lambda}(y_1, \dots, y_N), \quad (2.10)$$

where  $\lambda$  runs over all partitions with at most  $N$  parts. That is, we realize the Schur polynomial  $s_{\lambda}(x_1, \dots, x_N)$  as the partition function of the up-right model with  $N$  rows, weights  $W$ , and top boundary  $\lambda$ ; and we realize  $s_{\lambda}(y_1, \dots, y_N)$  as the partition function of the down-right model with  $N$  rows, weights  $W^*$ , and bottom boundary  $\lambda$ . See Figure 13 for an illustration.

**Remark 2.14.** The sums in the YBE (2.9) are finite for any fixed finite boundary  $i_1, i_2, i_3, j_1, j_2, j_3 \in \mathbb{Z}_{\geq 0}$  (see Figure 12), so the YBE is a polynomial identity in  $x, y$ . However, for the Cauchy sum (2.10) we employ the infinite boundary condition in the zeroth column, so the sum runs over all partitions (with at most  $N$  parts), and is a power series in the  $x_i, y_j$ , convergent for  $|x_i|, |y_j| < 1$ . Alternatively, it may be treated algebraically as a formal series. The two viewpoints are essentially equivalent, but since we are dealing with probabilistic connections, we adopt the analytic one.

We now explain how to use the YBE to evaluate the sum (2.10) as a finite product; the three steps are illustrated in Figure 14. Insert an auxiliary  $N \times N$  square of the cross-vertices  $\mathbb{R}_{x_i, y_j}$  to the right of the lattice in Figure 13. The crosses are initially empty thanks to the boundary

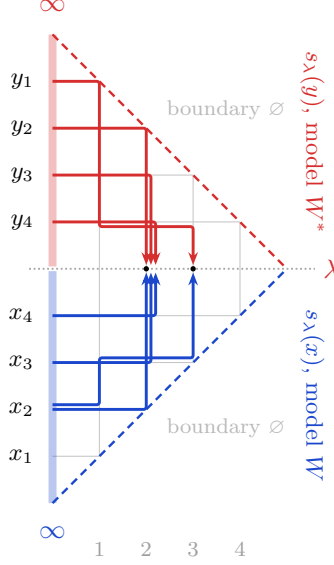


Figure 13: The vertex model for the Cauchy sum  $\sum_{\lambda} s_{\lambda}(x_1, \dots, x_N) s_{\lambda}(y_1, \dots, y_N)$  with  $N = 4$ .

conditions  $\emptyset$  on the right, so this inserted square must be empty, which contributes a factor of 1 to the partition function.

We then drag this square leftward through the lattice, one cross at a time, by the sequence of Yang–Baxter moves of Proposition 2.13 (see Figure 12). Each move preserves the partition function while altering the lattice by exchanging one blue line with one red. After  $N^2$  such moves, one for each pair  $x_i, y_j$ , all the crosses have passed to the left of the zeroth column, where they form a network fed by the empty left boundary (See Figure 14 for an illustration). Thanks to the boundary conditions  $\infty$  in the zeroth column, the outgoing occupations of each cross can be arbitrary, and the sum over them does not depend on the incoming ones:

$$\sum_{i_2, j_2 \geq 0} \mathbb{R}_{x_i, y_j}(i_1, j_1; i_2, j_2) = \sum_{a \geq 0} (x_i y_j)^a = \frac{1}{1 - x_i y_j} \quad \text{for every } (i_1, j_1), \quad (2.11)$$

because arrow conservation forces  $i_2 - j_2 = i_1 - j_1$ , leaving  $\min(i_2, j_2)$  to run over all of  $\mathbb{Z}_{\geq 0}$ . Summing the crosses in the reverse of the order in which they feed one another, each cross in turn has all of its outgoing edges free when it is summed, so the network contributes  $\prod_{i,j} (1 - x_i y_j)^{-1}$ . Only the first cross of the network is forced by the boundary to have incoming occupations  $(0, 0)$ ; the inner edges may carry nonzero occupations, as displayed in Figure 15. Meanwhile, the lattice to the right of the crosses, with the blue and red blocks interchanged, carries no paths at all (and thus has weight 1): an up-right path there would have to exit through the empty top boundary, and a down-right path through the empty bottom one. Thus, we obtain

**Proposition 2.15.** *For any  $N$  and  $x_1, \dots, x_N, y_1, \dots, y_N$  with  $|x_i|, |y_j| < 1$ , we have*

$$\sum_{\lambda} s_{\lambda}(x_1, \dots, x_N) s_{\lambda}(y_1, \dots, y_N) = \prod_{i,j=1}^N \frac{1}{1 - x_i y_j}, \quad (2.12)$$

where the sum is over all partitions  $\lambda$  with at most  $N$  parts.

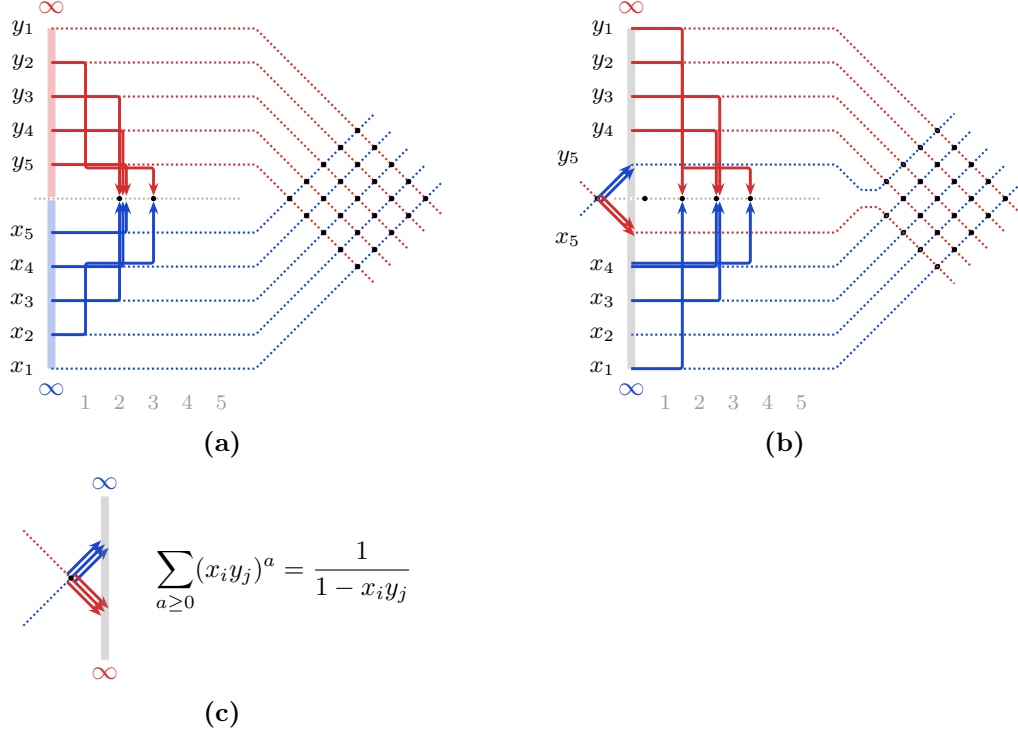


Figure 14: Illustration of the proof of the Cauchy identity (Proposition 2.15) by the YBE. (a) Attaching the empty  $N \times N$  square of  $\mathbb{R}_{x_i, y_j}$  vertices to the right of the lattice in Figure 13. (b) Dragging one cross through the lattice by the YBE. (c) After  $N^2$  moves, all crosses have reached the left boundary (see Figure 15 for a smaller but more detailed example). Each cross contributes a geometric series to the partition function.

## 2.6 Summary: Three kinds of Schur vertex models

The Schur polynomial has now appeared as the partition function of three vertex models: the five-vertex model (Section 2.2), the higher spin vertex model with capacity one of the horizontal edges (Remark 2.4), and the vertical-strip model (Section 2.4) with arbitrary capacities at the edges, but with the restriction like  $j_2 \leq i_1$ . Each of the three classes of models is a member of a hierarchy of solvable lattice models:

- The five-vertex model (Section 2.2) sits inside the *free fermion six-vertex model* (Remark 2.5, [ABPW23], [Nap24]). Furthermore, dropping the free fermion condition, we arrive at partition functions of general six-vertex models. Notably, this family also includes symmetric Grothendieck polynomials [MS13].
- The higher spin vertex model (Remark 2.4) was studied in connection with symmetric functions [Bor17], and then in connection with particle systems [CP16], [BP18]. This lifts the Schur polynomials to the Hall–Littlewood polynomials and further to the spin Hall–Littlewood symmetric rational functions.
- The vertical-strip model (Section 2.4) lifts the Schur polynomials to the  $q$ -Whittaker polynomials [Kor13, §6] and further to the spin  $q$ -Whittaker polynomials [BW21], [MP22], [BK24], [Kor24].

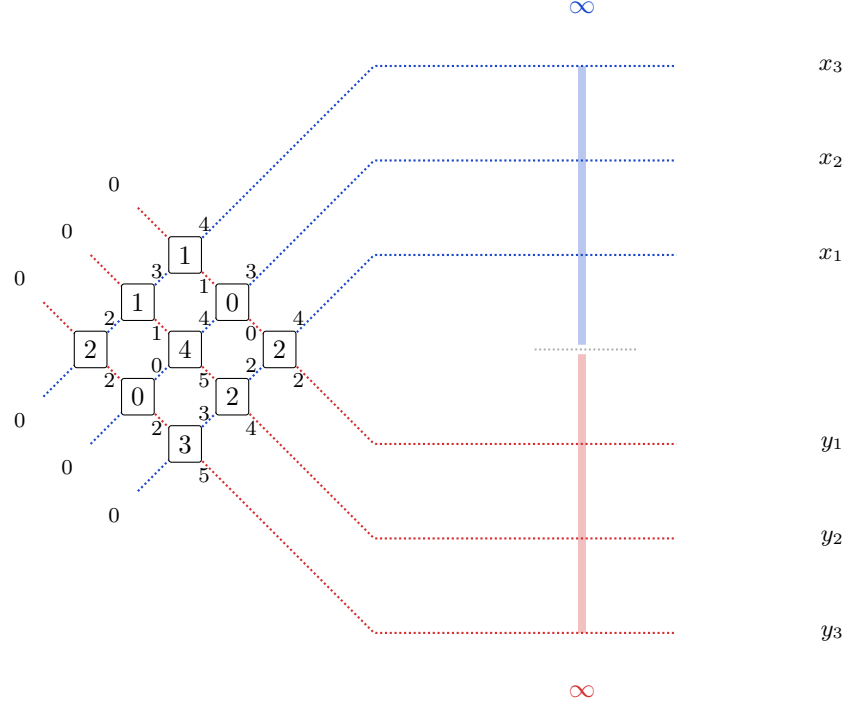


Figure 15: An example of a configuration in the final step of the Yang–Baxter moves in the proof of the Cauchy identity (Proposition 2.15). The numbers along the edges indicate occupations. The framed numbers on the vertices indicate  $\min(i_2, j_2)$ , which enter the weights  $\mathbb{R}_{x,y}$  (2.8). The edge occupations are uniquely determined by the framed numbers, thanks to the arrow preservation. The framed integers form a nonnegative integer matrix (the Robinson–Schensted–Knuth input matrix of Proposition 3.5), and the arrows recorded by the edge occupations are its Viennot shadow lines [Vie77], in the generalization from permutation matrices to arbitrary  $\mathbb{Z}_{\geq 0}$ -matrices [Pra15].

Despite the differences, these deformations are essentially instances of one general fully fused higher spin six-vertex model (connected to the quantum group  $U_q(\widehat{\mathfrak{sl}}_2)$ ). We will not describe this general model in full generality, but in Section 6 below we will discuss some of the deformations of the higher-spin and vertical-strip models, and connect them together via the fusion procedure.

### 3 Deriving RSK from the Yang–Baxter equation

We now match, term by term, the two sides of the YBE for the Schur vertex model that we used in Section 2.5. We will see that in this case the summands admit a unique weight-preserving matching, and identify it with the classical Robinson–Schensted–Knuth (RSK) row insertion algorithm. This material first appeared in [MP22, §5]. First, we recall the RSK in the toggle form.

#### 3.1 RSK via toggles

Classical RSK can be built entirely from local moves [Pak01], [Hop14], which we now explain. First, recall Fomin’s *growth-diagram* form of RSK [Fom88], [Fom95a]. Instead of building tableaux by insertion, one records the algorithm as a labeling of the vertices of an  $N \times N$  grid by partitions: the partitions at adjacent vertices interlace (that is, differ by a horizontal strip), and the nonnegative integer entry of the input matrix is placed inside the corresponding unit cell. The two output

tableaux  $P$  and  $Q$  are then read off the chains of partitions along the top and the right boundaries of the grid. See Figure 16 for an illustration of the growth diagram. A complete worked example — the insertion algorithm running in parallel with the corresponding vertex model states — is presented in Figure 21 at the end of this section.

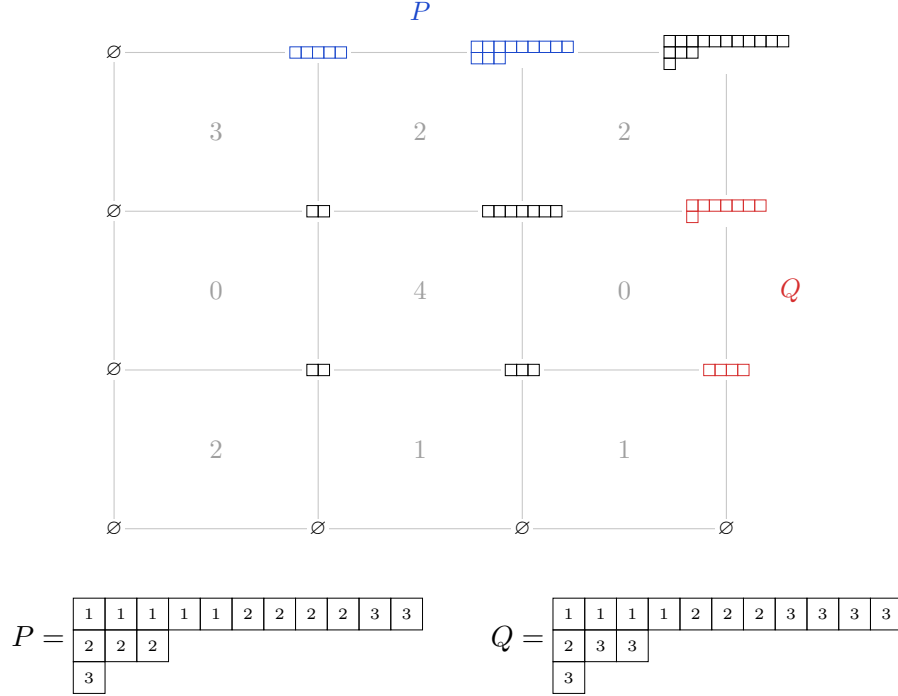
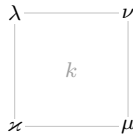


Figure 16: Fomin’s growth diagram. Each vertex carries a partition, drawn as its Young diagram; the left and bottom boundaries are empty ( $\emptyset$ ); adjacent partitions interlace by a horizontal strip. Each cell carries a nonnegative integer (the input matrix entry), and the top-right partition is determined by the other three corners and this integer via the local (toggle) rule (3.2). The top and right boundaries give the output Young tableaux  $P$  and  $Q$ , respectively.

The entire diagram is generated by a single *local (toggle) rule* that updates the top-right corner of a cell (denoted by  $\nu$ ) from the other three corners (denoted by  $\lambda, \mu, \varkappa$ ) and the integer inside the cell (denoted by  $k$ ):



The local rule assumes that  $\mu, \lambda$  are fixed, and it is a bijection between the following two sets:

$$\{\varkappa : \varkappa \prec \lambda, \varkappa \prec \mu\} \times \mathbb{Z}_{\geq 0} \longleftrightarrow \{\nu : \nu \succ \lambda, \nu \succ \mu\}, \quad (3.1)$$

which is defined by the “toggle” relations

$$\nu_1 = k + \max(\lambda_1, \mu_1), \quad \nu_{c+1} = \max(\lambda_{c+1}, \mu_{c+1}) + \min(\lambda_c, \mu_c) - \varkappa_c \quad (c \geq 1). \quad (3.2)$$

In Figure 20 at the end of this section, the roles of  $\varkappa, \lambda, \mu, \nu$  are marked in one cell of the growth diagram of the running example of Figure 21.

**Remark 3.1.** Note that other weight-compatible local rules — bijections between the two sets in (3.1) respecting the grading  $|\nu| = |\lambda| + |\mu| - |\kappa| + k$  — would correspond to other versions of the RSK correspondence, for example, the column insertion version. We refer to [Fom95a], [Fom95b] for further discussion.

### 3.2 The deterministic matching in the Schur case

The Schur YBE (2.9) for the weights (2.6), (2.7) and the cross-vertex weights (2.8) admits a *deterministic* term-by-term matching of the summands on its two sides. First, note that, as in any YBE of the form (2.9), for fixed boundary conditions occupations  $i_1, i_2, i_3, j_1, j_2, j_3 \in \mathbb{Z}_{\geq 0}$ , both sides of (2.9) vanish unless the global balance  $i_1 + j_2 + j_3 = i_2 + i_3 + j_1$  holds. Under this balance condition, path conservation at the three vertices leaves a single free internal coordinate on each side,

$$(k_1, k_2, k_3) = (k_1, i_2 + k_1 - i_1, i_3 + j_1 - k_1), \quad (k'_1, k'_2, k'_3) = (k'_1, k'_1 + j_2 - j_1, j_3 + i_1 - k'_1), \quad (3.3)$$

respectively.

**Lemma 3.2.** *In the Yang–Baxter equation (2.9) for the Schur vertex model, with boundary occupations satisfying the balance  $i_1 + j_2 + j_3 = i_2 + i_3 + j_1$ , the summands in the left- and right-hand sides are, respectively,*

$$x^{j_2} y^{j_1} (xy)^{\min(k_1, k_2)} \quad \text{and} \quad x^{k'_2} y^{k'_1} (xy)^{\min(j_1, j_2)}. \quad (3.4)$$

*These are nonzero exactly on the integer intervals*

$$\max(0, i_1 - i_2) \leq k_1 \leq \min(i_3, i_3 + j_1 - j_2), \quad \max(0, j_1 - j_2) \leq k'_1 \leq \min(j_3, j_3 + i_1 - i_2), \quad (3.5)$$

*which contain the same number of points.*

*Moreover, the map*

$$k'_1 = j_1 - \min(j_1, j_2) + \min(k_1, k_2) = k_1 + \max(0, j_1 - j_2) - \max(0, i_1 - i_2) \quad (3.6)$$

*is a weight-preserving bijection between the weights (3.4), and it is the only one: within each of the two lists (3.4) the monomials are pairwise distinct, so a weight-preserving matching of the two lists has no freedom.*

*Proof.* The formulas (3.4) are obtained by substituting (3.3) into the vertex weights (2.6), (2.7), and (2.8). The occupation bounds  $k_1, k_2 \geq 0$  and  $k_3 \geq \max(j_1, j_2)$  in (2.6), (2.7) (and their primed counterparts) cut out the two desired intervals (3.5). Finally, note that since  $k'_1 + \min(j_1, j_2) = j_1 + \min(k_1, k_2)$  and  $k'_2 = k'_1 + j_2 - j_1$ , the map (3.6) turns the right-hand summand  $x^{k'_2} y^{k'_1} (xy)^{\min(j_1, j_2)}$  into the left-hand one,  $x^{j_2} y^{j_1} (xy)^{\min(k_1, k_2)}$ . Moreover, it is a bijection between the two intervals (3.5). For the uniqueness, observe that  $\min(k_1, k_2) = k_1 + \min(0, i_2 - i_1)$  and  $k'_2 = k'_1 + j_2 - j_1$ , so the exponent of  $xy$  in the left-hand summand and the exponent of  $y$  in the right-hand one are strictly increasing in  $k_1$  and  $k'_1$ , respectively. Hence the monomials within each list are pairwise distinct, and a weight-preserving bijection between the lists is unique. This completes the proof.  $\square$

**Remark 3.3.** As discussed in the Introduction, the rigidity in Lemma 3.2 is special to the Schur weights. Section 5 develops the general coupling of the two sides of a YBE by Markov transitions; the matching (3.6) is its deterministic case, the one with  $\mathbf{p}^{\text{fwd}}, \mathbf{p}^{\text{bwd}} \in \{0, 1\}$  in (5.2).

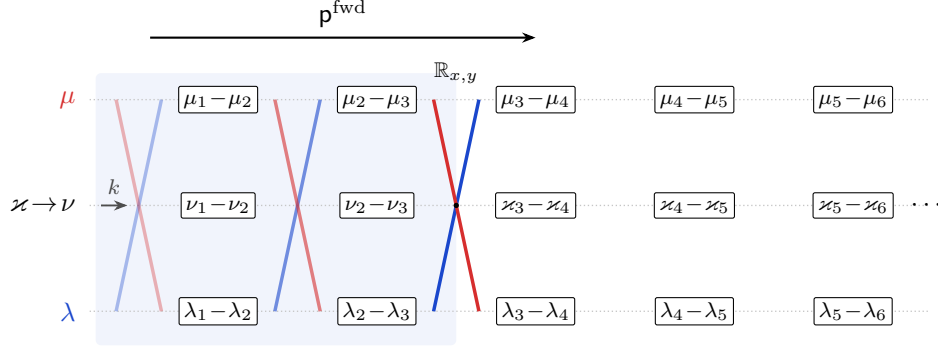


Figure 17: Dragging the Yang–Baxter cross through two rows  $\mu/\kappa$  (top) and  $\lambda/\kappa$  (bottom) of a growth diagram. Each vertical edge carries a column multiplicity  $\square_c - \square_{c+1}$ . The middle row is rewritten  $\kappa \mapsto \nu$  (the shaded columns are already updated). The only free input is  $k \in \mathbb{Z}_{\geq 0}$  at the first column, which corresponds to the state of the cross vertex  $\mathbb{R}_{x,y}$  to the left of the zeroth column (the simplest example is in Figure 14, (c)). We discuss this free input in detail at the end of the proof of Proposition 3.5. The arrow marks the direction of the drag; its label  $\mathbf{p}^{\text{fwd}}$  is the forward transition probability of Section 5, which for the Schur weights takes only the values 0 and 1.

**Remark 3.4.** The bijection (3.6) takes a cleaner form in terms of the vertical arrow counts  $k_3 = i_3 + j_1 - k_1$ ,  $k'_3 = j_3 + i_1 - k'_1$ :

$$k'_3 = k_3 + \max(i_1, i_2) - \max(j_1, j_2). \quad (3.7)$$

**Proposition 3.5.** *Dragging the cross (2.8) rightward across the two rows  $\lambda/\kappa$  and  $\mu/\kappa$  of Figure 17, and rewriting the middle rail  $\kappa \mapsto \nu$  one column at a time by the deterministic move of Lemma 3.2, reproduces the local rule (3.2). Consequently, the Yang–Baxter moves over the  $N \times N$  grid (Figure 16) produce the Robinson–Schensted–Knuth correspondence.<sup>2</sup>*

*Proof.* Fix a column  $c \geq 1$ . The three rails give the vertical occupations

$$i_3 = \lambda_c - \lambda_{c+1}, \quad j_3 = \mu_c - \mu_{c+1}, \quad k_3 = \kappa_c - \kappa_{c+1}, \quad k'_3 = \nu_c - \nu_{c+1}, \quad (3.8)$$

the middle edge reading  $\kappa$  ahead of the cross and  $\nu$  behind it. Each horizontal leg carries the number of paths crossing it, a partial sum of the strip multiplicities: the two legs entering the cross carry  $i_1 = \nu_c - \mu_c$  and  $i_2 = \nu_c - \lambda_c$ , and the two leaving it carry  $j_1 = \lambda_{c+1} - \kappa_{c+1}$  and  $j_2 = \mu_{c+1} - \kappa_{c+1}$ , all nonnegative since  $\kappa \prec \lambda$ ,  $\kappa \prec \mu$ ,  $\lambda \prec \nu$ , and  $\mu \prec \nu$ . Hence

$$\max(i_1, i_2) = \nu_c - \min(\lambda_c, \mu_c), \quad \max(j_1, j_2) = \max(\lambda_{c+1}, \mu_{c+1}) - \kappa_{c+1}. \quad (3.9)$$

Substituting (3.8)–(3.9) into the  $k_3$ -identity (3.7) and cancelling  $\nu_c$  and  $\kappa_{c+1}$  gives

$$\nu_{c+1} = \max(\lambda_{c+1}, \mu_{c+1}) + \min(\lambda_c, \mu_c) - \kappa_c,$$

which is precisely the toggle (3.2) for  $c \geq 1$ .

The case  $c = 0$  is slightly different, because here each Yang–Baxter cross carries the free input (an element of the integer input matrix for the RSK). This input matrix is encoded by the  $N \times N$

<sup>2</sup>This RSK correspondence is the most classical one, based on row insertions. It would be interesting to see if the other three bijections (column insertion, dual row insertion, dual column insertion) can be derived from the Yang–Baxter equation in a similar way.

grid as in Figure 16. Assume that we need to insert the cross  $\mathbb{R}_{x_a, y_b}$ ,  $1 \leq a, b \leq N$ , through the zeroth column of the grid which carries infinitely many vertical paths. Let this cross carry the parameter  $k_{ab} \in \mathbb{Z}_{\geq 0}$ . Assume that the input arrows to  $\mathbb{R}_{x_a, y_b}$  are  $i_1$  and  $i_2$ , see Figure 18, and  $k_{ab} = \min(k_1, k_2)$ . This means that

$$k_1 = k_{ab} + \max(0, i_1 - i_2), \quad k_2 = k_{ab} + \max(0, i_2 - i_1).$$

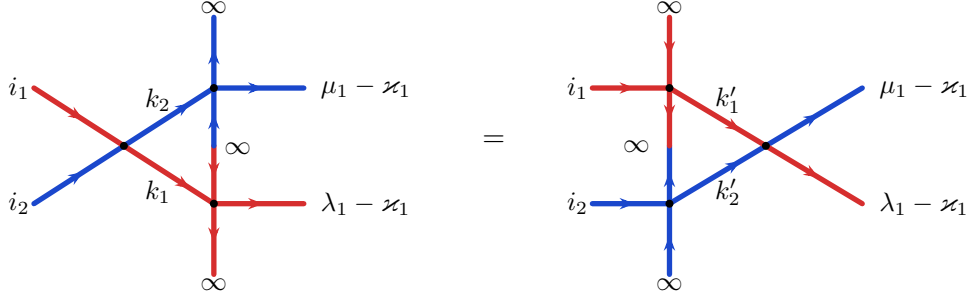


Figure 18: Dragging the cross through the zeroth column.

After dragging the cross, we have  $k'_1 = \nu_1 - \mu_1$  and  $k'_2 = \nu_1 - \lambda_1$ . Comparing the weights on the left and on the right in Figure 18, we see that

$$\nu_1 = k_{ab} + \max(\lambda_1, \mu_1),$$

which does not depend on the incoming occupations  $i_1, i_2$ .

Thus, the data  $k_{ab} = \min(k_1, k_2)$  at each cross  $\mathbb{R}_{x_a, y_b}$  in Figure 15 is exactly the RSK input matrix. This completes the proof of Proposition 3.5.  $\square$

Figure 19 fixes the input matrix  $M$  of the running example and its biword. Figure 21 illustrates the resulting correspondence step by step, with the bumping cascades on the RSK side matched to the path hops on the seam of the vertex model. Figure 20 displays the growth diagram of the same input matrix.

$$M = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 1 & 0 \end{pmatrix} \longrightarrow \left( \begin{array}{cc|cc|cc} 1 & 1 & 2 & 2 & 3 & 3 \\ 0 & 2 & 2 & 2 & 1 & 2 \\ 1 & 1 & 0 & 0 & 0 & 0 \end{array} \right) \begin{array}{l} b \text{ --- recorded in } Q \\ a \text{ --- inserted into } P \end{array}$$

entry  $m_{ba}$ : row  $b$ , column  $a$  biword:  $m_{ba}$  copies of  $\begin{pmatrix} b \\ a \end{pmatrix}$ , in lexicographic order

Figure 19: The input matrix  $M$  of the running example and its biword. The top row of the biword (the  $b$ 's, red) is recorded in  $Q$ , and the bottom row (the  $a$ 's, blue) is inserted into  $P$ .

## 4 The combinatorial 3D $R$ and the tetrahedron equation

The matching of Lemma 3.2 is the only weight-preserving bijection between the summands of the two sides of (2.9). A reader who has met *set-theoretic* solutions of the Yang–Baxter equation will ask whether this matching is one of them. In this section we answer this question. As a map on the arrow occupations of a single cross, the Schur matching is not a binary Yang–Baxter map at all. In three coordinates read off the monomial weights, however, it becomes the standard *combinatorial*

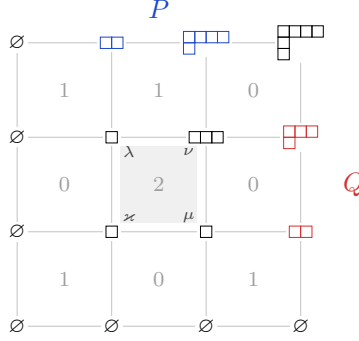


Figure 20: The growth diagram for the  $3 \times 3$  input matrix of Figure 19; the top and right boundary chains are the  $P$ - and  $Q$ -chains. In the shaded cell, the corners play the roles  $\kappa = (1)$ ,  $\lambda = (1)$ ,  $\mu = (1)$  in the local rule (3.2) with  $k = 2$ , which yields  $\nu = (3)$ .

*three-dimensional*  $R$ , a map of triples whose consistency equation is the Zamolodchikov tetrahedron equation — one dimension above the Yang–Baxter equation (2.9). Moreover, we also explain how ordinary set-theoretic Yang–Baxter maps reappear from this map of triples after a periodic closure, see Section 4.3.3 below.

#### 4.1 Is the Schur matching a set-theoretic Yang–Baxter map?

We refer to [RV05] and [KP11] for background on set-theoretic Yang–Baxter maps which are defined as follows.

**Definition 4.1.** Let  $X$  be a set. A *set-theoretic Yang–Baxter map* is a map  $r: X \times X \rightarrow X \times X$  satisfying the braid relation

$$r_{12} r_{23} r_{12} = r_{23} r_{12} r_{23} \quad \text{on } X^3, \quad (4.1)$$

where  $r_{ab}$  applies  $r$  to the factors  $a$  and  $b$  and acts as the identity on the remaining one.

**Remark 4.2.** The references above write (4.1) in the “quantum form”  $R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}$ . The braid form (4.1) used here corresponds to it via  $r = P \circ R$ , where  $P$  is the flip of the two factors.

**Definition 4.3.** There exists a colored version of Definition 4.1, in which one is given sets  $X_l$  and maps  $r_{l,m}: X_l \times X_m \rightarrow X_m \times X_l$  satisfying the evident analogue of (4.1) on  $X_k \times X_l \times X_m$ .

There are two natural candidates for a binary map coming out of the Schur matching described in Section 3. Neither of them is a set-theoretic Yang–Baxter map:

- The cross vertex  $\mathbb{R}_{x,y}$  itself, read as a map from its two incoming occupations to its two outgoing ones is not a function, since it assigns weights to multiple outgoing states for a single incoming state.
- The deterministic matching (3.6) of Lemma 3.2, which sends the free internal occupation  $k_1$  on the left-hand side of the YBE to the free internal occupation  $k'_1$  on the right-hand side, for fixed boundary occupations. It is a bijection for each fixed boundary, but not a map on edge states.

However, we do not claim that no other packaging of the Schur matching as a set-theoretic Yang–Baxter map exists. Instead, in the next Section 4.2 we explain how the Schur matching possesses a related three-dimensional structure.

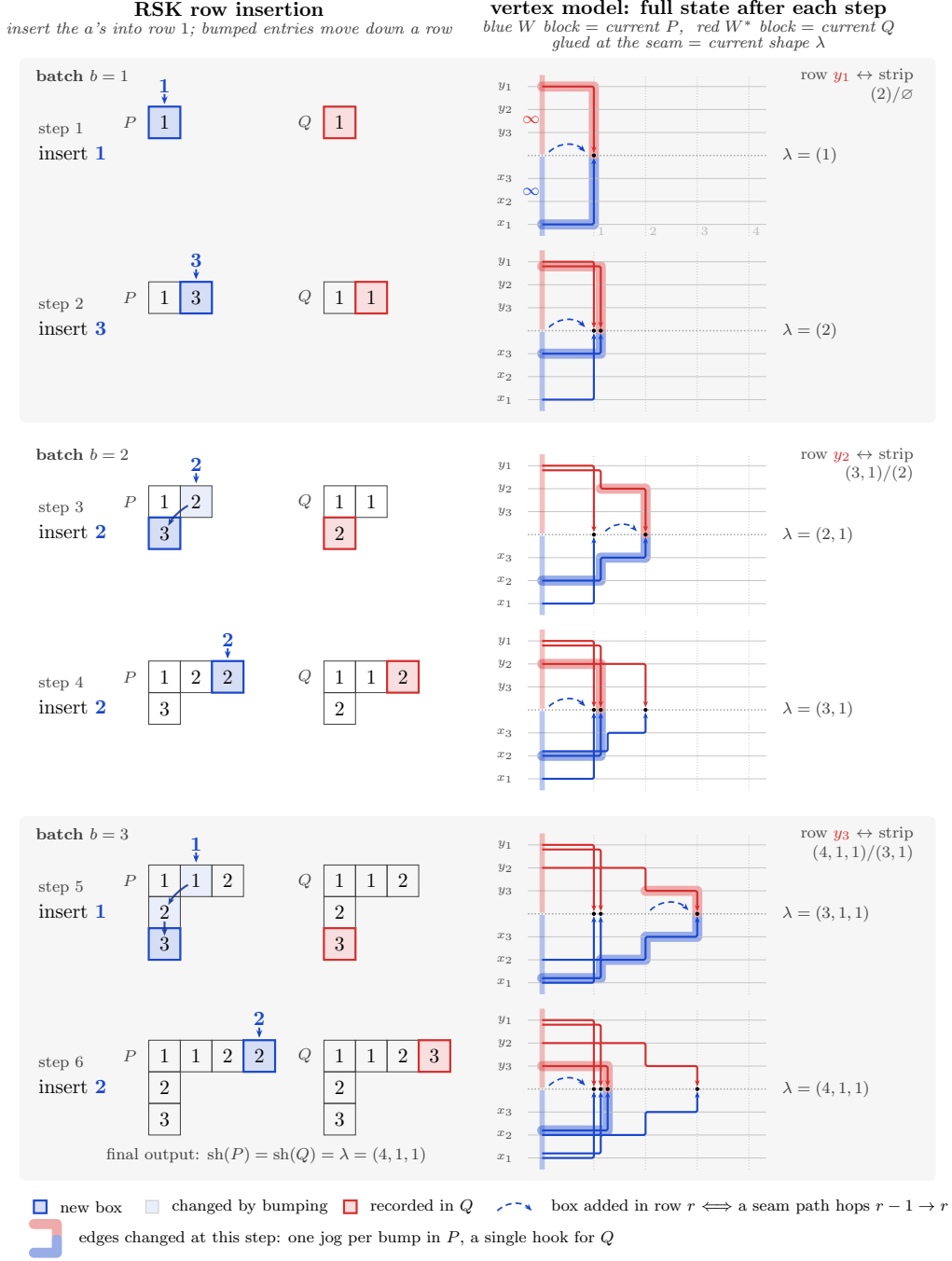


Figure 21: RSK insertion and the Schur vertex model running in parallel on the input matrix  $M$  of Figure 19. After each step, the blue block (rows  $x_a$ , weights  $W$  (2.6)) encodes the current  $P$ , and the red block (rows  $y_b$ , weights  $W^*$  (2.7)) encodes the current  $Q$ . The blocks are glued along the seam carrying the common shape  $\lambda$  in the column coordinates (2.5). A new box in row  $r$  of  $\lambda$  moves one seam path from column  $r - 1$  to column  $r$  (dashed arcs). Shaded corridors mark the edges that change at each step — the trail of the cross  $\mathbb{R}_{x_a, y_b}$  being dragged in the proof of Proposition 3.5: the blue corridor enters at row  $x_a$  and jogs once per entry bumped in  $P$ , while the red corridor is a single hook at row  $y_b$ , as  $Q$  only gains one box.

## 4.2 The hidden three-dimensional $R$

The three coordinates to keep for the desired three-dimensional structure are the ones that carry the monomial exponents in (3.4). Set

$$a = j_1, \quad h = \min(k_1, k_2), \quad c = j_2, \quad m = \min(j_1, j_2) = \min(a, c), \quad (4.2)$$

so that the left-hand summand is  $x^c y^a (xy)^h$  and the right-hand one is  $x^{k'_2} y^{k'_1} (xy)^m$ . In these coordinates the first equality in (3.6) reads  $k'_1 = a - m + h = h + \max(a - c, 0)$ , and the conservation relation  $k'_2 = k'_1 + j_2 - j_1$  then gives  $k'_2 = h + c - m = h + \max(c - a, 0)$ . Both outputs are thus expressed through  $(a, h, c)$  alone, by the map

$$R^{(3)}(a, h, c) = (h + \max(a - c, 0), \min(a, c), h + \max(c - a, 0)), \quad R^{(3)}: \mathbb{Z}_{\geq 0}^3 \rightarrow \mathbb{Z}_{\geq 0}^3. \quad (4.3)$$

This map is the standard combinatorial three-dimensional  $R$  of Kuniba–Okado [KO12] and Kuniba [Kun16].

In the coordinates (4.2), weight preservation in Lemma 3.2 reads

$$a + h = k'_1 + m, \quad h + c = m + k'_2, \quad (4.4)$$

the first matching the exponents of  $y$  in (3.4) and the second those of  $x$ . Thus  $R^{(3)}$  preserves each of the finite sets

$$C_{p,q} = \{(i, j, k) \in \mathbb{Z}_{\geq 0}^3 : i + j = p, j + k = q\}, \quad p, q \in \mathbb{Z}_{\geq 0}, \quad (4.5)$$

on which the middle coordinate  $j$  ranges over  $0 \leq j \leq \min(p, q)$  and determines the other two. On  $C_{p,q}$  the map  $R^{(3)}$  is the reversal  $j \mapsto \min(p, q) - j$  of the middle coordinate, since  $j' = \min(i, k) = \min(p, q) - j$ . In particular,  $R^{(3)}$  is an involution.

A map of triples has its own consistency equation, one step above (4.1).

**Definition 4.4.** Let  $X$  be a set. A *set-theoretic tetrahedron map* is a map  $R: X^3 \rightarrow X^3$  satisfying the Zamolodchikov tetrahedron equation [Zam81]

$$R_{123} R_{145} R_{246} R_{356} = R_{356} R_{246} R_{145} R_{123} \quad \text{on } X^6, \quad (4.6)$$

where  $R_{abc}$  applies  $R$  to the factors  $a, b, c$  and acts as the identity on the other three. The four index triples 123, 145, 246, 356 are the four faces of a tetrahedron, each pair of them meeting in exactly one index; see Figure 22.<sup>3</sup>

For a fixed boundary,  $h$  recovers the original internal state through

$$k_1 = h + \max(i_1 - i_2, 0), \quad k_2 = h + \max(i_2 - i_1, 0), \quad k_3 = i_3 + j_1 - k_1, \quad k'_3 = j_3 + i_1 - k'_1, \quad (4.7)$$

the first two by (3.3) and the definition of  $h$ , and the last two by (3.3) again. See Figure 23.

Thus, we arrive at the following statement, which is the main result of this section.

**Proposition 4.5.** *Let  $h = \min(k_1, k_2)$  and  $m = \min(j_1, j_2)$ .*

(i) *Under the deterministic Schur matching (3.6), the three active occupations transform as*

$$(j_1, h, j_2) \mapsto (k'_1, m, k'_2) = R^{(3)}(j_1, h, j_2). \quad (4.8)$$

---

<sup>3</sup>Note that some sources use the indexing 124, 135, 236, 456, obtained by interchanging the labels 3 and 4, and conventions for composing maps may reverse both sides of (4.6).

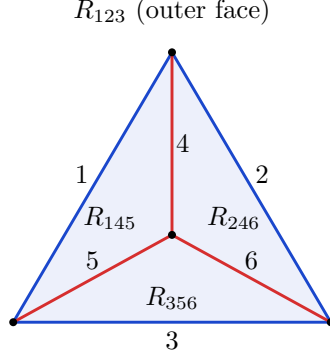


Figure 22: The four factors of the tetrahedron equation (4.6). The six tensor factors of  $X^6$  are drawn as the six edges of a tetrahedron, and each  $R_{abc}$  acts on the three edges of one face: 123 is the outer face, and 145, 246, 356 are the three faces meeting at the interior vertex.

- (ii) *Consequently, after adjoining the fixed boundary data  $(i_1, i_2, i_3, j_1, j_2, j_3)$  needed to reconstruct the remaining arrows, the Schur matching is the restriction of  $R^{(3)}$  to a finite boundary-dependent slice of  $\mathbb{Z}_{\geq 0}^3$ .*
- (iii) ([KO12], [Kun16]) *The map  $R^{(3)}$  is an involutive set-theoretic tetrahedron map: it satisfies (4.6) on  $\mathbb{Z}_{\geq 0}^6$ .*
- (iv) *Conversely, every triple  $(a, h, c) \in \mathbb{Z}_{\geq 0}^3$  arises as the active triple of the Schur matching for some boundary. Thus  $R^{(3)}$  on all of  $\mathbb{Z}_{\geq 0}^3$  is assembled from the Schur matchings over the various boundaries.*

*Proof.* Part (i) is the computation preceding (4.3). For (ii), fix a boundary. Then  $a = j_1$  and  $c = j_2$  are frozen and only  $h$  varies, over the image of the interval (3.5) under  $k_1 \mapsto \min(k_1, k_2)$ . The slice is therefore the one-dimensional set  $\{(j_1, h, j_2)\}$  with  $h$  running over that image. Part (iii) is a theorem of the cited literature, and we do not reproduce its proof. For (iv), given  $(a, h, c) \in \mathbb{Z}_{\geq 0}^3$ , take the boundary

$$i_1 = i_2 = 0, \quad i_3 = h + \max(c - a, 0), \quad j_1 = a, \quad j_2 = c, \quad j_3 = h + \max(a - c, 0).$$

The global balance  $i_1 + j_2 + j_3 = i_2 + i_3 + j_1$  holds, since both sides equal  $h + \max(a, c)$ . The state  $k_1 = k_2 = h$  lies in the interval (3.5), and (4.7) gives  $k_3 = \max(a, c)$  and  $k'_3 = 0$ .  $\square$

The combinatorial three-dimensional  $R$ , its conserved sectors, its involutivity, and the tetrahedron equation (4.6) come from the quantized-function-algebra and crystal literature. The closest references for the form used here are Kuniba–Okado [KO12] and Kuniba [Kun16]. The latter also contains the periodic reductions (Section 4.3.3 below) and the local relations (4.10) below. For the quantum three-dimensional  $R$  and for dimensional reduction, see Kuniba–Okado [KO15].

The map  $R^{(3)}$  is the  $q \rightarrow 0$  limit of the quantum three-dimensional  $R$  [KO12, Remark 2.6] and, up to reversing the two outer coordinates, the transition map between the two Lusztig (PBW) parametrizations for the type  $A_2$  braid move, see Lusztig [Lus90, p. 451] and [KO12, §2.5].

Schur polynomials also arise from  $q$ -oscillator solutions of the tetrahedron equation in Iwao–Motegi–Ohkawa [IMO25], [IMO26], but that construction does not involve RSK. Igonin–Konstantinou–Rizos [IKR23] show in general that solving a local Yang–Baxter correspondence produces set-theoretic tetrahedron maps. Gleizer–Postnikov [GP00] found a Yang–Baxter type relation for

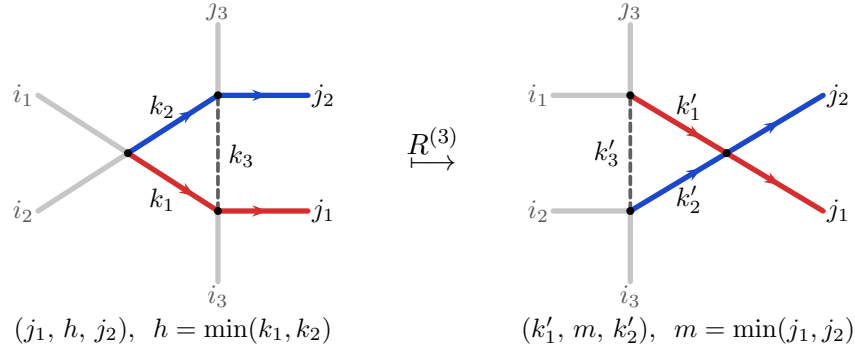


Figure 23: The active part (4.8) of the Yang–Baxter move of Figure 12. The matching of Lemma 3.2 moves the triple  $(j_1, h, j_2)$  to the triple  $(k'_1, m, k'_2)$  by the combinatorial three-dimensional  $R$  of (4.3). The gray edges  $i_1, i_2, i_3, j_3$  are spectators: they are the same on the two sides. The dashed middle segment of the vertical line is not a spectator — it changes from  $k_3$  to  $k'_3$  — but both values are recovered from the active triple and the boundary through (4.7).

scattering matrices describing tensor products of  $GL(N)$  representations, whose piecewise-linear transformation of parameters solves the tetrahedron equation. Genuine set-theoretic Yang–Baxter maps built from insertion include the plactic braiding of Lebed [Leb20] and the combinatorial  $R$ -matrices on rectangular tableaux of Shimozono [Shi02] and Frieden [Fri21]. Both use global insertion data rather than the fixed-boundary matching of Lemma 3.2.

We have not found in the literature the identification (4.8) of the active part of the Schur Yang–Baxter matching with the combinatorial three-dimensional  $R$ , nor the comparison between its periodic reduction and the open RSK train that we describe in Section 4.3.1 below.

### 4.3 From three-dimensional $R$ to RSK, row insertion, and box-ball systems

The local map (4.3) acts on a single triple. To obtain a map on pairs of sequences  $\mathbf{u}, \mathbf{v} \in \mathbb{Z}_{\geq 0}^n$  (the column-difference sequences  $(\lambda_t - \lambda_{t+1})_t$  of partitions, as in the dictionary (3.8)), one composes  $n$  copies of it along a line of sites, and one must choose which of the three coordinates is the carrier, that is, which output of one site is fed as the input of the next site, in which direction the sites are visited, and whether the carrier is closed periodically or left open with fixed boundary data. Since  $R^{(3)}$  is invariant under the exchange of its first and third coordinates, the carrier is either an outer coordinate or the middle one, and with the two boundary conditions this leaves four cases. All of these four combinations lead to known objects, as we explain in this subsection. We begin with the case that has already appeared in Section 3 above.

#### 4.3.1 Middle coordinate, open: RSK

In Section 3, the carrier is the middle coordinate, left open: we drag the cross through two rows with fixed boundary data, as in Figure 17. Write  $h_c = \min(\lambda_c, \mu_c) - \varkappa_c$  for  $c \geq 1$ , and let  $h_0 = k_{ab}$  be the free input at the zeroth column. The dictionary (3.8) turns the move at column  $c$  into

$$R^{(3)}(\lambda_{c+1} - \varkappa_{c+1}, h_c, \mu_{c+1} - \varkappa_{c+1}) = (\nu_{c+1} - \mu_{c+1}, h_{c+1}, \nu_{c+1} - \lambda_{c+1}), \quad c \geq 0. \quad (4.9)$$

For  $c \geq 1$  this is Proposition 4.5, with  $h_c$  the middle coordinate  $\min(k_1, k_2)$  of (4.2). For  $c = 0$ , where the column carries infinitely many vertical paths and the proposition does not apply, it is the computation of  $\nu_1$  in the proof of Proposition 3.5. The free input  $h_0 = k_{ab}$  is the  $(a, b)$ -entry

of the RSK input matrix, arrow conservation turns the moves at the later columns into the toggle (3.2), and iterating over the grid gives the growth diagram, hence RSK.

### 4.3.2 Third coordinate, open: row insertion

Taking the third coordinate as the carrier, visiting the sites in the direction of increasing  $t$  and leaving the carrier open, gives the Schensted row insertion. Indeed, let  $\mathbf{u}, \mathbf{v} \in \mathbb{Z}_{\geq 0}^n$  record the letter counts of two one-row tableaux, and solve

$$(a_t, b_t, c_{t+1}) = R^{(3)}(u_t, v_t, c_t), \quad t = 1, \dots, n, \quad c_1 = 0.$$

Then  $\mathbf{a}$  and  $\mathbf{b}$  are the letter counts of the first and second rows of the tableau obtained by row inserting the letters of  $\mathbf{v}$  into  $\mathbf{u}$ , and  $c_t$  is the number of letters  $< t$  by which the new first row exceeds  $\mathbf{u}$ . See Figure 24 for an example. Running the carrier in the direction of decreasing  $t$  instead amounts to reversing the alphabet.

The map  $(\mathbf{u}, \mathbf{v}) \mapsto (\mathbf{b}, \mathbf{a})$  is the plactic braiding of Lebed [Leb20], an idempotent, non-invertible solution of (4.1) on  $X = \mathbb{Z}_{\geq 0}^n$ . In a birational setting, Noumi–Yamada [NY04] obtain both tropical row insertion and a birational counterpart of the combinatorial  $R$ -matrix from a system of discrete Toda type, with open and quasi-periodic indexing, respectively.

$$\begin{array}{ccc} \mathbf{u} = (2, 1, 0) & \boxed{1 \mid 1 \mid 2} & \xrightarrow{\text{row insertion of } \mathbf{v} \text{ into } \mathbf{u}} \begin{array}{c} \boxed{1 \mid 1 \mid 1 \mid 3} \\ \boxed{2} \end{array} & \begin{array}{l} \mathbf{a} = (3, 0, 1) \\ \mathbf{b} = (0, 1, 0) \end{array} \\ \mathbf{v} = (1, 0, 1) & \boxed{1 \mid 3} & & \end{array}$$

Figure 24: Row insertion in letter counts, with  $n = 3$ . The one-row tableaux  $112$  and  $13$  have letter counts  $\mathbf{u}$  and  $\mathbf{v}$ ; inserting the letters of  $\mathbf{v}$  into the first row bumps a  $2$  into the second row, and the two rows of the result have letter counts  $\mathbf{a}$  and  $\mathbf{b}$ . The carrier is  $(c_1, c_2, c_3, c_4) = (0, 1, 0, 1)$ : the new first row has one more letter  $1$  than  $\mathbf{u}$ , the same number of letters  $\leq 2$ , and one more letter  $\leq 3$ .

### 4.3.3 Third coordinate, periodic: combinatorial $R$ -matrices and the box-ball system

Fix  $n \geq 1$  and, for  $l \geq 1$ , let  $B_l^{(n)} = \{(u_1, \dots, u_n) \in \mathbb{Z}_{\geq 0}^n : u_1 + \dots + u_n = l\}$ , the one-row symmetric-tensor crystal. Given  $\mathbf{u} \in B_l^{(n)}$  and  $\mathbf{v} \in B_{l'}^{(n)}$ , take the third coordinate as the carrier, visit the sites in the direction of decreasing  $t$ , and close the carrier periodically: look for carrier occupations  $c_1, \dots, c_n$  with  $c_0 = c_n$  such that

$$(a_t, b_t, c_{t-1}) = R^{(3)}(u_t, v_t, c_t), \quad t = 1, \dots, n, \quad (4.10)$$

that is,  $a_t = v_t + \max(u_t - c_t, 0)$ ,  $b_t = \min(u_t, c_t)$  and  $c_{t-1} = v_t + \max(c_t - u_t, 0)$ . The recursion determines the carrier from  $c_n$ , so this is one equation for one unknown. For  $l > l'$  it has a unique solution, and for  $l = l'$  every sufficiently large  $c_n$  solves it with the same output. Setting  $\mathcal{R}_{l,l'}(\mathbf{u}, \mathbf{v}) = (\mathbf{b}, \mathbf{a})$  in these cases and  $\mathcal{R}_{l,l'} := (\mathcal{R}_{l',l})^{-1}$  for  $l < l'$  gives bijections

$$\mathcal{R}_{l,l'}: B_l^{(n)} \times B_{l'}^{(n)} \rightarrow B_{l'}^{(n)} \times B_l^{(n)}.$$

The tetrahedron equation (4.6) for the local map implies the colored set-theoretic Yang–Baxter relation of Definition 4.3 for the family  $\{\mathcal{R}_{l,l'}\}$ . These  $\mathcal{R}_{l,l'}$  are the classical combinatorial  $R$ -matrices of the crystals  $B_l^{(n)}$ , going back to Nakayashiki–Yamada [NY97], and the time evolution

of the box-ball system is their successive application, with a carrier passing through a row of boxes. The relations (4.10), their solvability, and this identification are due to [Kun16, §6].

**Remark 4.6.** Box-ball systems and RSK are also linked directly in another manner, without passing through  $R^{(3)}$ : the recent work of Imamura–Mucciconi–Sasamoto–Scrimshaw [IMSS26] reads Fomin’s local rule for Schensted column insertion as a two-lane box-ball system with a carrier, and projects it onto the classical box-ball system.

#### 4.3.4 Middle coordinate, periodic: the combinatorial $R$ -matrix with a dual crystal

Taking the middle coordinate as the carrier, visiting the sites in the direction of increasing  $t$  and closing the carrier periodically, is simpler, because the middle output  $\min(a, c)$  of (4.3) does not depend on the middle input  $h$ . For  $\mathbf{u} \in B_l^{(n)}$  and  $\mathbf{v} \in B_{l'}^{(n)}$ , the relations

$$(a_t, c_t, b_t) = R^{(3)}(u_t, c_{t+1}, v_t), \quad t = 1, \dots, n, \quad c_{n+1} = c_1,$$

force  $c_t = \min(u_t, v_t)$  and give  $a_t = u_t + \delta_t$ ,  $b_t = v_t + \delta_t$  with  $\delta_t = \min(u_{t+1}, v_{t+1}) - \min(u_t, v_t)$ . The map  $(\mathbf{u}, \mathbf{v}) \mapsto (\mathbf{b}, \mathbf{a})$  is the combinatorial  $R$ -matrix  $R^\vee$  of Kuniba–Okado–Yamada [KOY05]. Like  $\mathcal{R}_{l,l'}$ , it is a set-theoretic Yang–Baxter map and not a three-dimensional one, but it pairs  $B_l^{(n)}$  with the dual crystal, and drives their box-ball system with a reflecting end. They also record the Yang–Baxter relations satisfied by  $R^\vee$  together with the maps  $\mathcal{R}_{l,l'}$  of Section 4.3.3. Trace reductions of the tetrahedron equation, including those along the middle coordinate, are treated in [Kun22, Chapter 13].

## 5 Bijectivization: From identities to Markov transitions

We outline the idea of *bijectivization* (probabilistic bijection), the device that converts the algebraic Yang–Baxter identities of the previous sections into stochastic (Markov) steps.

When both sides are sums of nonnegative weights,  $\sum_a \text{wt}(a) = \sum_b \text{wt}(b)$ , we *randomize* the identity by introducing a transformation — a Markov step — which turns a configuration on the left into a randomly chosen configuration on the right. This randomization must be consistent with the weights, but its choice need not be unique. Carried across the lattice, these random steps are interpreted as a random growth process. We call this passage from an identity to a randomized transformation a *bijectivization* or a *probabilistic bijection*. It is an application, in a solvable lattice setting, of the general principle of *coupling* in probability theory; see the books of Lindvall [Lin02] and Thorisson [Tho00].

Bijectivization of the Yang–Baxter equation was introduced in [BP19], extending the earlier works which constructed randomized (Markov) RSK-like insertion processes for Hall–Littlewood and  $q$ -Whittaker polynomials [OP13], [BP16], [MP17], [BP15], [BBW16]. In integrable probability, Yang–Baxter bijectivization has been applied further to connect spin Hall–Littlewood and spin  $q$ -Whittaker polynomials [BMP21], [MP22] to various stochastic vertex models; to construct Markov maps acting on the fixed-time distributions of TASEP [PS22], [PS24] or preserving the KPZ pure states of the stochastic six-vertex model on the full plane [NP23]. Furthermore, bijectivization has been extended to half-space models [CD21] and to the full two-parameter Macdonald setting [AF22], [FSA24].

Strikingly, the same mechanism appears independently in two-dimensional critical statistical mechanics in the proof of rotational invariance of the Ising, random-cluster, and six-vertex models. Namely, the star-triangle (Yang–Baxter) relation is realized as a move that “uses extra randomness”

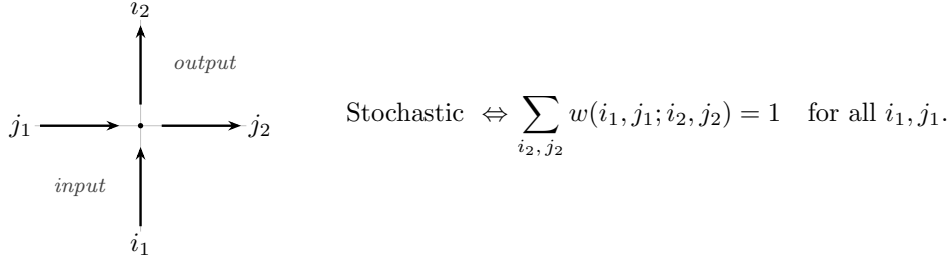


Figure 25: A single vertex has an *input* (the incoming occupations  $(i_1, j_1)$  on the bottom and left edges) and an *output* (the outgoing occupations  $(i_2, j_2)$  on the top and right). The Boltzmann weight of the vertex  $w(i_1, j_1; i_2, j_2)$  is called *stochastic* when these weights are the conditional probabilities of the output given the input.

and “is not a deterministic matching of the different configurations” [DCKK<sup>+</sup>20]. This is the general mechanism which we are about to describe.

Consider any identity with positive terms,

$$\sum_{a \in A} \text{wt}(a) = \sum_{b \in B} \text{wt}(b), \quad (5.1)$$

where  $A, B$  are nonempty finite or countable index sets,  $\text{wt} > 0$ , and the common total mass  $Z := \sum_{a \in A} \text{wt}(a) = \sum_{b \in B} \text{wt}(b)$  is finite. A *bijectivization* (or probabilistic bijection) of (5.1) is a pair of nonnegative matrices  $\mathbf{p}^{\text{fwd}}(a \rightarrow b)$  and  $\mathbf{p}^{\text{bwd}}(b \rightarrow a)$ , indexed by  $a \in A$  and  $b \in B$ , satisfying the stochasticity

$$\sum_{b \in B} \mathbf{p}^{\text{fwd}}(a \rightarrow b) = 1, \quad \sum_{a \in A} \mathbf{p}^{\text{bwd}}(b \rightarrow a) = 1,$$

and the reversibility (detailed balance) relation

$$\mathbf{p}^{\text{fwd}}(a \rightarrow b) \text{wt}(a) = \mathbf{p}^{\text{bwd}}(b \rightarrow a) \text{wt}(b) \quad \text{for all } a \in A, b \in B. \quad (5.2)$$

Summing (5.2) over  $b$  recovers each term  $\text{wt}(a)$  of the left side of (5.1) from  $\mathbf{p}^{\text{fwd}}$  being stochastic, and summing over  $a$  recovers each term of the right side; thus a bijectivization is exactly a *coupling* [Lin02], [Tho00] of the two sides of the identity. That is, the joint measure on  $A \times B$  with weights  $\mathbf{p}^{\text{fwd}}(a \rightarrow b) \text{wt}(a)$  has marginals  $\text{wt}(a)$  and  $\text{wt}(b)$  on  $A$  and  $B$ , respectively; dividing it by  $Z$  turns it into a probability distribution.

Let us highlight two extreme cases of bijectivization:

- If both  $\mathbf{p}^{\text{fwd}}$  and  $\mathbf{p}^{\text{bwd}}$  are equal to zero or one, then they are supported on the graphs of two mutually inverse maps, and the coupling is an honest weight-preserving *bijection*  $a \mapsto b$ . (A zero-one  $\mathbf{p}^{\text{fwd}}$  by itself only makes the transport deterministic: it may send several  $a$ ’s to one  $b$ , and then  $\mathbf{p}^{\text{bwd}}$  is genuinely random.) One can say that this bijection provides a combinatorial proof of (5.1).
- The opposite extreme is the *independent* (“maximally random”) bijectivization

$$\mathbf{p}^{\text{fwd}}(a \rightarrow b) = \frac{\text{wt}(b)}{\sum_{b' \in B} \text{wt}(b')}, \quad \mathbf{p}^{\text{bwd}}(b \rightarrow a) = \frac{\text{wt}(a)}{\sum_{a' \in A} \text{wt}(a')},$$

which makes the output independent of the input. The joint measure in this case is  $Z^{-1} \text{wt}(a) \text{wt}(b)$ ; dividing it by  $Z$  as above turns it into the product of the two normalized marginals  $\text{wt}(a)/Z$  and  $\text{wt}(b)/Z$ .

A bijectivization always exists, as the second example shows. It is unique only when one of  $A, B$  is a singleton. When both  $A$  and  $B$  have exactly two elements, the bijectivization is a one-parameter family.

When bijectivization is applied to the YBE (2.9), the sets  $A$  and  $B$  are the sets of arrow configurations of the internal edges (indexed by  $k_1, k_2, k_3$  and  $k'_1, k'_2, k'_3$ , respectively) which carry a nonzero weight, and the weights are the products of the three vertex weights on each side. Here and below,  $A$  and  $B$  always mean the supports of the two sums: configurations of weight zero are discarded, and all stochasticity and uniqueness statements are imposed on the positive-weight support only. The full power of bijectivization is realized when it is applied sequentially on the lattice, by dragging the cross vertex through several columns, one at a time. In particular, if the cross is dragged from left to right, then locality of the moves implies that the distribution of  $k'_1, k'_2, k'_3$  in the first (leftmost) column does not depend on the subsequent columns.

For the Schur weights this construction admits the first of the two extreme cases above: the matching (3.6) of Lemma 3.2 is a bijectivization of (2.9) with  $p^{\text{fwd}}, p^{\text{bwd}} \in \{0, 1\}$ , and dragging the cross across the lattice then produces the deterministic RSK correspondence of Proposition 3.5 instead of a random dynamics. For generic parameter values, the  $q$ - and  $t$ -deformed models of Section 6 have boundary data admitting no termwise matching of the two sides, and the couplings they require are random; see Section 7.

## 6 $q$ -deformations: Hall–Littlewood, $q$ -Whittaker, and fusion

The  $t$ - and  $q$ -deformations treated in this section continue the probabilistic half of these notes, begun in Section 5.

Putting the five-vertex model aside, we now focus on the two other Schur models of Section 2. Each of them carries deformations that preserve the YBE and hence the Cauchy identity (proven exactly in the same way as in Section 2.5). We do not explore the full generality of the deformations (detailed in [Bor17], [BP18] and [BW21], [MP22], [BK24], [Kor24]), but focus on the symmetric-function cases of Hall–Littlewood and  $q$ -Whittaker polynomials.

### 6.1 Vertex model for Hall–Littlewood polynomials

The Hall–Littlewood weights are the  $t$ -deformation of the higher-spin six-vertex model of Remark 2.4: horizontal edges carry  $j_1, j_2 \in \{0, 1\}$ , the vertical edges have unbounded capacity, and arrows are conserved,  $i_1 + j_1 = i_2 + j_2$ . The four nonzero weights, with spectral parameter  $x$ , are

$$\begin{array}{cccc}
 \begin{array}{c} \text{---} \\ \uparrow \uparrow \uparrow \\ \text{---} \end{array} & \begin{array}{c} \text{---} \\ \uparrow \uparrow \uparrow \rightarrow \\ \text{---} \end{array} & \begin{array}{c} \text{---} \\ \uparrow \uparrow \uparrow \\ \text{---} \end{array} & \begin{array}{c} \text{---} \\ \uparrow \uparrow \uparrow \rightarrow \\ \text{---} \end{array} \\
 w = 1 & w = x & w = 1 & w = x(1 - t^g)
 \end{array} \tag{6.1}$$

where  $g$  denotes the occupation of the incoming (bottom) vertical edge ( $g = 3$  in the pictures), and all other configurations are weighted 0. At  $t = 0$ , we recover the Schur weights from Figure 8. Up to the horizontal and vertical gauge factors  $x^{j_2 - j_1}$  and  $(t; t)_{i_1} / (t; t)_{i_2}$ , respectively, these weights are the Fock space matrix elements of the  $q$ -boson  $L$ -operator of [BIK98]; see [Kor13, §3.2].

**Proposition 6.1** ([Tsi06], [Kor13], [FW09], [Bor17]). *For a partition  $\lambda$  with  $\ell(\lambda) \leq N$ , the partition function of the higher-spin model with  $N$  rows of spectral parameters  $x_1, \dots, x_N$ , weights (6.1), top boundary  $\lambda$  (given by the multiset  $\{\lambda_1, \dots, \lambda_N\}$ ), fully packed left boundary, and empty bottom and right boundaries (see Figure 8) equals the Hall–Littlewood polynomial  $P_\lambda(x_1, \dots, x_N; 0, t)$ .*

A standard reference on Hall–Littlewood and Macdonald polynomials is the book [Mac95]. Here and below, we use the notation  $P_\lambda(\cdots; q, t)$  for the Macdonald polynomials. For  $q = 0$ , they reduce to the Hall–Littlewood polynomials. For  $t = 0$ , Macdonald polynomials become the  $q$ -Whittaker polynomials, which we discuss in Section 6.2 below. Both families become Schur polynomials at  $q = t$  (which, in applications to vertex models, is taken to be  $q = t = 0$ ).

**Remark 6.2** (Spin Hall–Littlewood). Adjoining a *spin* parameter  $s$  deforms the four weights (6.1), in the same order, into

$$\frac{1 - sxt^g}{1 - sx}, \quad \frac{x - st^g}{1 - sx}, \quad \frac{1 - s^2t^g}{1 - sx}, \quad \frac{x(1 - t^g)}{1 - sx},$$

which return (6.1) at  $s = 0$ . Their partition functions are, up to normalization, the *spin Hall–Littlewood* rational functions [Bor17], [BP18]. One may even let  $s \rightsquigarrow s_j$  vary column by column without breaking the YBE.

## 6.2 Vertex model for $q$ -Whittaker polynomials

Now consider a  $q$ -deformation of the up-right weight  $W$  (2.6), whose edges carry arbitrary multiplicities  $i_1, j_1, i_2, j_2 \in \mathbb{Z}_{\geq 0}$ , subject to the conservation law  $i_1 + j_1 = i_2 + j_2$  and the restriction  $j_2 \leq i_1$ . The  $q$ -deformed weight is defined as

$$W(i_1, j_1; i_2, j_2) = \mathbf{1}_{i_1+j_1=i_2+j_2} \mathbf{1}_{i_1 \geq j_2} x^{j_2} \frac{(q; q)_{i_1}}{(q; q)_{j_2} (q; q)_{i_1-j_2}}, \quad (6.2)$$

where  $(a; q)_m = \prod_{k=0}^{m-1} (1 - aq^k)$  is the  $q$ -Pochhammer symbol (with  $(a; q)_0 = 1$ ,  $(a; q)_m = \prod_{k=0}^{m-1} (1 - aq^{m+k})^{-1}$  for  $m < 0$ , and  $(a; q)_\infty = \prod_{k \geq 0} (1 - aq^k)$ ). The down-right weight, deforming (2.7), is

$$W^*(i_1, j_1; i_2, j_2) = \mathbf{1}_{i_2+j_1=i_1+j_2} \mathbf{1}_{i_2 \geq j_2} x^{j_2} \frac{(q; q)_{i_1}}{(q; q)_{j_2} (q; q)_{i_2-j_2}}. \quad (6.3)$$

As  $q \rightarrow 0$ , both weights reduce to (2.6) and (2.7). Unlike in the Schur case, (6.3) is not the plain reflection  $i_1 \leftrightarrow i_2$  of (6.2), but carries an extra vertical-edge factor  $(q; q)_{i_1}/(q; q)_{i_2}$ . It telescopes along a column, and it is required for the YBE of Section 7.1.<sup>4</sup>

**Remark 6.3.** The ratio  $(q; q)_{i_1}/(q; q)_{j_2} (q; q)_{i_1-j_2}$  is the  $q$ -binomial coefficient  $\begin{bmatrix} i_1 \\ j_2 \end{bmatrix}_q$ , a standard  $q$ -deformation of the binomial coefficient  $\binom{i_1}{j_2}$ . Note that the “classical” limit for us is  $q \rightarrow 0$ , not  $q \rightarrow 1$ .

**Proposition 6.4** (cf. [Kor13, §6.2]). *For a partition  $\lambda$ , the partition function of the up-right model with  $N$  rows of spectral parameters  $x_1, \dots, x_N$ , weights (6.2), top boundary  $\lambda$ , empty bottom boundary, and the infinite column-0 reservoir of Remark 2.11 on the left (see Figure 11) equals the  $q$ -Whittaker polynomial  $Q_\lambda(x_1, \dots, x_N; q, 0) = b_\lambda P_\lambda(x_1, \dots, x_N; q, 0)$ , where  $b_\lambda = \prod_{i \geq 1} (q; q)_{\lambda_i - \lambda_{i+1}}^{-1}$  does not depend on the  $x_i$ ’s. The down-right weights (6.3), with the mirrored boundary conditions, give  $P_\lambda$  itself. At  $q = 0$  one has  $b_\lambda = 1$ .*

<sup>4</sup>The asymmetry also reflects the shape of the Cauchy identity, which pairs  $P_\lambda$  with  $Q_\lambda$  rather than with a second copy of  $P_\lambda$ . The two families are merely proportional,  $Q_\lambda = b_\lambda P_\lambda$  with  $b_\lambda = \prod_{i \geq 1} (q; q)_{\lambda_i - \lambda_{i+1}}^{-1}$  depending on  $\lambda$  and  $q$  but not on the variables. The two models are normalized accordingly: the up-right model (6.2) produces  $Q_\lambda$  (Proposition 6.4), while the telescoped factor in (6.3) makes the down-right model produce  $P_\lambda$ .

**Remark 6.5.** The column-0 reservoir vertex carries a  $q$ -Pochhammer factor: as  $M \rightarrow \infty$ ,  $\left[\frac{M}{j}\right]_q \rightarrow 1/(q; q)_j$ , so that  $W(\infty, j_1; \infty, j_2) = x^{j_2}/(q; q)_{j_2}$  (the  $q$ -analogue of Remark 2.11). For  $N = 1$  and  $\lambda = (m)$  this single vertex already gives  $x^m/(q; q)_m = Q_{(m)}(x; q, 0)$ , whereas  $P_{(m)}(x; q, 0) = x^m$ . For general  $N$  the factor  $b_\lambda$  is a global normalization, see Remark 6.7 below.

**Remark 6.6.** The weights (6.2) were introduced in [Kor13, §6.1] (up to a left–right reflection of the lattice, and with  $q$  denoted by  $t$  there), as Fock space matrix elements of a  $q$ -boson  $L$ -operator, together with the identification of the partition functions with the skew  $q$ -Whittaker functions — in the cylindric (periodic) setting of that paper, which reduces to Proposition 6.4 for the empty bottom boundary. The YBE established in [Kor13, §3.2] exchanges two rows of the same model via an infinite-dimensional cross depending on the ratio of the two spectral parameters, and yields the symmetry of the partition functions in the  $x_i$ ’s.

The spin form of the weights (6.2) is already present in [Bor17, §6], as the fused weights at their  $q$ -Hahn point, together with a Cauchy identity pairing them with the Hall–Littlewood side — obtained there by analytic continuation in the fusion parameter  $q^J$ , without a YBE for the fused weights themselves. The dual weights (6.3) and the YBE of Cauchy type, in the form of an exchange relation between row operators, first appeared, in spin generality, in [BW21, §4.6, §5.9] (with the resulting Cauchy identity in §7.1 there); the explicit cross (7.2) was computed, in its spin form, in [BMP21, App. A].

**Remark 6.7** (Spin and diagonal parameters). One can add a spin parameter  $s$  further deforming (6.2) into the spin  $q$ -Whittaker weight of [BW21], [MP22]:

$$W(i_1, j_1; i_2, j_2) = \mathbf{1}_{i_1+j_1=i_2+j_2} \mathbf{1}_{i_1 \geq j_2} x^{j_2} \frac{(-s/x; q)_{j_2} (-sx; q)_{i_1-j_2} (q; q)_{i_2}}{(q; q)_{j_2} (q; q)_{i_1-j_2} (s^2; q)_{i_2}}.$$

At  $s = 0$  this is  $(q; q)_{i_2}/(q; q)_{i_1}$  times (6.2), not (6.2) itself; the discrepancy is a vertical-edge gauge, which telescopes along each column to the overall factor  $b_\lambda^{-1}$ , so the  $s = 0$  spin partition function equals  $P_\lambda$  rather than the  $Q_\lambda = b_\lambda P_\lambda$  of Proposition 6.4. The corresponding partition functions, with suitably normalized reservoir weights, are the spin  $q$ -Whittaker polynomials (unlike the spin Hall–Littlewood symmetric rational functions, these are polynomials in the  $x_i$ ’s). Going further, [BK24], [Kor24] enriched the spin  $q$ -Whittaker weights with extra *diagonal* parameters carried by the vertices; these have a representation-theoretic meaning and interpolation properties.

### 6.3 Macdonald duality

Here we illustrate how one can turn Hall–Littlewood polynomials into the  $q$ -Whittaker ones, and connect this construction to vertex models. First, consider the Cauchy identity for Hall–Littlewood polynomials [Mac95, Ch. III, §4]:<sup>5</sup>

$$\sum_{\lambda} P_{\lambda}(x_1, \dots, x_N; 0, t) Q_{\lambda}(y_1, \dots, y_M; 0, t) = \prod_{i=1}^N \prod_{j=1}^M \frac{1 - t x_i y_j}{1 - x_i y_j}. \quad (6.4)$$

Here,  $Q_{\lambda}$  is a multiple of  $P_{\lambda}$ , where the coefficient depends on  $t$  and  $\lambda$  but not on the  $y_j$ ’s.

We now perform the following steps with (6.4):

- Replace each  $y_i$  by a finite geometric sequence  $y_i, t y_i, \dots, t^{J-1} y_i$  of some length  $J \in \mathbb{Z}_{\geq 1}$  (denote this finite geometric sequence by  $y_i^{(J)}$ ). This is a finite version of the *plethystic substitution* appearing in symmetric function theory. On the vertex model side, this corresponds

<sup>5</sup>This Cauchy identity also admits a Yang–Baxter proof, e.g., see [Bor17].

to *fusion* of  $J$  rows of the Hall–Littlewood vertex model, and dates back to [KRS81]. This turns the Cauchy identity (6.4) into

$$\sum_{\lambda} P_{\lambda}(x_1, \dots, x_N; 0, t) Q_{\lambda}(y_1^{(J)}, \dots, y_M^{(J)}; 0, t) = \prod_{i=1}^N \prod_{j=1}^M \frac{1 - t^J x_i y_j}{1 - x_i y_j}. \quad (6.5)$$

- Both sides of (6.5) depend analytically on the  $2M$  parameters  $y_i$  and  $t^J y_i$ , treated as independent, in a neighborhood of  $y_i = 0$ . Indeed, each  $Q_{\lambda}(\cdots; 0, t)$  is a polynomial in the power sums, and the substitution  $y_i \mapsto y_i^{(J)}$  turns the  $k$ -th power sum into  $\sum_{i=1}^M (y_i^k - (t^J y_i)^k)/(1 - t^k)$ . The same dependence on the pair  $x, t^J x$  appears in the fused weights (6.11) of Section 6.4 below. In particular, we may perform the substitutions

$$y_i \rightsquigarrow 0, \quad t^J y_i \rightsquigarrow -z_i, \quad i = 1, \dots, M,$$

where  $z_i$  are new indeterminates. The Cauchy identity (6.5) becomes

$$\sum_{\lambda} P_{\lambda}(x_1, \dots, x_N; 0, t) F_{\lambda}(z_1, \dots, z_M) = \prod_{i=1}^N \prod_{j=1}^M (1 + x_i z_j). \quad (6.6)$$

- It remains to identify the symmetric functions  $F_{\lambda}(z_1, \dots, z_M)$  in (6.6). Thanks to the Macdonald duality  $\omega_{q,t}$  which interchanges  $q$  and  $t$  in the Macdonald polynomials (and transposes the partitions, turning  $Q_{\lambda}$  into  $P_{\lambda'}$ ), the Hall–Littlewood and  $q$ -Whittaker polynomials satisfy the following dual Cauchy identity:

$$\sum_{\lambda} P_{\lambda}(x_1, \dots, x_N; 0, t) P_{\lambda'}(z_1, \dots, z_M; t, 0) = \prod_{i=1}^N \prod_{j=1}^M (1 + x_i z_j). \quad (6.7)$$

Since, for each  $N$ , the identity (6.7) determines the functions  $P_{\lambda'}(\cdots; t, 0)$  with  $\ell(\lambda) \leq N$  (by the linear independence of the polynomials  $P_{\lambda}(x_1, \dots, x_N; 0, t)$  with  $\ell(\lambda) \leq N$ ), and  $N$  is arbitrary, we conclude that

$$F_{\lambda}(z_1, \dots, z_M) = P_{\lambda'}(z_1, \dots, z_M; t, 0),$$

which are exactly the  $q$ -Whittaker polynomials (where the role of  $q$  is taken by the parameter  $t$ ).

## 6.4 Fusion

The same steps as in Section 6.3 may be carried out directly at the level of the vertex weights. Let us briefly outline this procedure, which is known as *fusion* [KRS81], [KR87], [Bor17], [CP16], [BP18].

The fundamental building block is the Hall–Littlewood vertex (6.1), of horizontal capacity one, so  $j_1, j_2 \in \{0, 1\}$ . Fix  $J \in \mathbb{Z}_{\geq 1}$  and consider  $J$  copies of the vertex, stacked vertically. Assign to the  $r$ -th row the spectral parameter  $t^{r-1}x$ , so that the column carries the geometric string of parameters. This mirrors the substitution  $y_i \mapsto y_i^{(J)}$  from Section 6.3 above.

The fused weight, with  $i_1, i_2 \in \mathbb{Z}_{\geq 0}$  and  $0 \leq j_1, j_2 \leq J$ , is defined as the sum over the  $J$  incoming and outgoing horizontal occupations  $\vec{h}_1, \vec{h}_2 \in \{0, 1\}^J$ :

$$w_x^{(J)}(i_1, j_1; i_2, j_2) = \frac{\mathbf{1}_{i_1+j_1=i_2+j_2}}{Z_{j_1}(J)} \sum_{\substack{|\vec{h}_1|=j_1 \\ |\vec{h}_2|=j_2}} t^{\sum_{r=1}^J (r-1) h_1^{(r)}} \prod_{r=1}^J w_{t^{r-1}x}(\ell_{r-1}, h_1^{(r)}; \ell_r, h_2^{(r)}), \quad (6.8)$$

where  $w_u$  are the weights (6.1) with spectral parameter  $u$ , and the internal vertical occupations are determined by arrow conservation,  $\ell_0 = i_1$  and  $\ell_r = \ell_{r-1} + h_1^{(r)} - h_2^{(r)}$ ; the indicator is what makes  $\ell_J = i_2$ . Strings that leave  $\mathbb{Z}_{\geq 0}$  need not be excluded by hand: the first step with  $\ell_r = -1$  is the fourth vertex of (6.1) at  $g = 0$ , of weight  $x(1 - t^0) = 0$ . See Figure 26 for an illustration of the fusion procedure.

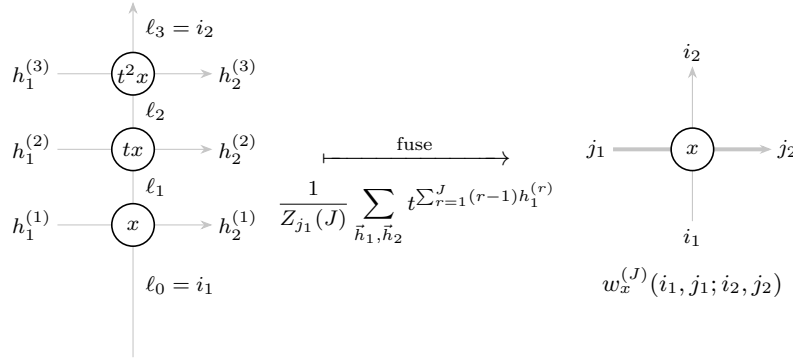


Figure 26: The fusion procedure (6.8) for  $J = 3$ . Summing over the incoming and outgoing strings  $\vec{h}_1, \vec{h}_2 \in \{0, 1\}^J$  with totals  $j_1 = |\vec{h}_1|$  and  $j_2 = |\vec{h}_2|$ , with the  $t$ -exchangeable weight on the incoming arrows, collapses the column to a single fused vertex of horizontal capacity  $J$ .

The normalization in (6.8) is the  $t$ -deformed binomial

$$Z_j(J) = \sum_{\vec{h}_1: |\vec{h}_1|=j} t^{\sum_{r=1}^J (r-1) h_1^{(r)}} = t^{\binom{j}{2}} \left[ \begin{matrix} J \\ j \end{matrix} \right]_t.$$

The weight  $t^{\sum_{r=1}^J (r-1) h_1^{(r)}} / Z_{j_1}(J)$  placed on the incoming arrows is the  $t$ -exchangeable normalization (e.g., see [GO10] for a probabilistic perspective). Fusion is built so as to preserve the YBE [KRS81], [KR87]; the fused weights (6.8) therefore satisfy it as well (with a correspondingly fused cross), as do their stochastic counterparts [Bor17], [BP18]. By construction (6.8) is a genuine vertex weight of horizontal capacity  $J$  — a function of the four occupations  $i_1, j_1, i_2, j_2$  alone.

In fact, the outer sum over the outgoing string in (6.8) is redundant: a fused column maps  $t$ -exchangeable inputs to  $t$ -exchangeable outputs [CP16, Prop. 3.6], so that for *every* fixed  $\vec{h}_2$  with  $|\vec{h}_2| = j_2$ , we have the following refined identity:

$$\sum_{\vec{h}_1: |\vec{h}_1|=j_1} t^{\sum_{r=1}^J (r-1) h_1^{(r)}} \prod_{r=1}^J w_{t^{r-1}x}(\ell_{r-1}, h_1^{(r)}; \ell_r, h_2^{(r)}) = \frac{Z_{j_1}(J)}{Z_{j_2}(J)} t^{\sum_{r=1}^J (r-1) h_2^{(r)}} w_x^{(J)}(i_1, j_1; i_2, j_2). \quad (6.9)$$

<sup>6</sup>Throughout this subsection we assume  $0 < t < 1$ , so that  $Z_j(J) > 0$  for all  $0 \leq j \leq J$ . At  $t = 0$  the normalization vanishes for  $j \geq 2$  and (6.8) reads  $0/0$ ; the fused weight is then defined by continuity, through the closed form (6.11) below, whose coefficients  $c_p(t)$  are regular at  $t = 0$ .

**Remark 6.8.** Let us illustrate (6.9) for  $J = 2$ . The general- $J$  case of (6.9) follows from the same computation applied to adjacent rows, since the symmetrization underlying  $t$ -exchangeability is generated by swaps of neighboring rows.

Take  $i_1 = 0, j_1 = 1$ , so the incoming string is  $(1, 0)$  with weight 1 and  $(0, 1)$  with weight  $t$ . For the outgoing string  $\vec{h}_2 = (0, 1)$  the two incoming terms give

$$\underbrace{t w_x(0, 0; 0, 0) w_{tx}(0, 1; 0, 1)}_{= t^2 x} + \underbrace{w_x(0, 1; 1, 0) w_{tx}(1, 0; 0, 1)}_{= tx(1-t)} = tx = x t^1,$$

while  $\vec{h}_2 = (1, 0)$  gives  $x = x t^0$ . Since  $j_1 = j_2 = 1$  makes  $Z_{j_1}/Z_{j_2} = 1$  in (6.9), either outgoing string already yields  $w_x^{(2)}(0, 1; 0, 1) = x$ , with no outer sum needed.

Carrying out the fusion sum (6.8) — a direct if lengthy computation — gives the fused weight in closed form:<sup>7</sup>

$$\begin{aligned} w_x^{(J)}(i_1, j_1; i_2, j_2) &= \mathbf{1}_{i_1+j_1=i_2+j_2} t^{\frac{1}{2}i_1(i_1+2j_1-1)+\frac{1}{2}(j_1-i_2)(j_1-i_2-1)} x^{j_2} \frac{(t; t)_{j_1}}{(t; t)_{j_2} (t; t)_{i_2}} \\ &\quad \times \sum_{k=0}^{i_1} \frac{(t^{-i_1}; t)_k (t^{-i_2}; t)_k}{(t; t)_k} (t^{1+j_1-i_2+k}; t)_{i_1-k} (t^{1+J-i_1-j_1+k}; t)_{i_1-k} t^k. \end{aligned} \quad (6.10)$$

Since the prefactor multiplying  $x^{j_2}$  is free of both  $x$  and  $J$ , we may write, for  $i_1 + j_1 = i_2 + j_2$ ,  $w_x^{(J)}(i_1, j_1; i_2, j_2) = x^{j_2} \sum_{p=0}^{i_1} c_p(t) (t^J)^p$ . In particular  $t^J$  may be set to any complex value.

Expanding the one  $t^J$ -dependent factor by the  $t$ -binomial theorem and evaluating the inner sum by the terminating  $t$ -Chu–Vandermonde identity [GR04, Eq. (II.6)] gives  $c_p(t)$  explicitly:

$$c_p(t) = (-1)^p t^{(j_2-p)} \frac{(t; t)_{j_1}}{(t; t)_{j_2} (t; t)_{i_2}} \begin{bmatrix} i_1 \\ p \end{bmatrix}_t (t^{1+j_2-p}; t)_p (t^{1+j_1}; t)_{i_1-p}, \quad 0 \leq p \leq i_1.$$

The factor  $(t^{1+j_2-p}; t)_p$  acquires the vanishing term  $1 - t^0$  as soon as  $p > j_2$ , so we see that the degree of (6.10) in  $t^J$  is equal to  $\min(i_1, j_2)$ .

Put  $z = -t^J x$ . Then

$$w_x^{(J)}(i_1, j_1; i_2, j_2) = x^{j_2} \sum_{p=0}^{\min(i_1, j_2)} c_p(t) (-z/x)^p = \sum_{p=0}^{\min(i_1, j_2)} c_p(t) (-1)^p x^{j_2-p} z^p. \quad (6.11)$$

Send  $x \rightarrow 0$  while keeping  $z$  fixed. If  $i_1 < j_2$ , then (6.11) vanishes, which leads to the indicator  $\mathbf{1}_{i_1 \geq j_2}$  in (6.2). If  $i_1 \geq j_2$ , then the limit picks only the term with  $p = j_2$ , giving the  $q$ -Whittaker weight:

$$\mathbf{1}_{i_1 \geq j_2} z^{j_2} \begin{bmatrix} i_1 \\ j_2 \end{bmatrix}_t = \mathbf{1}_{i_1 \geq j_2} z^{j_2} \frac{(t; t)_{i_1}}{(t; t)_{j_2} (t; t)_{i_1-j_2}},$$

where the parameter  $t$  plays the role of  $q$  in (6.2).

**Remark 6.9** (Spin parameter). Carrying the spin parameter  $s$  of Remark 6.2 through the fusion works equally well, and produces the *spin*  $q$ -Whittaker weights of Remark 6.7 — but through a different specialization: not  $x \rightarrow 0$  with  $z = -t^J x$  fixed, but  $x = s$ , so that  $t^J = -z/s$

<sup>7</sup>This computation may be reduced to checking certain recurrences for  $w_x^{(J)}$ . We refer to [CP16, Lem. 3.13, Thm. 3.15] for a detailed derivation via these recurrences in the presence of the additional spin parameter.

[BMP21, App. A.3], followed by the vertical-edge gauge  $(t; t)_{i_2}(s^2; t)_{i_1}/[(t; t)_{i_1}(s^2; t)_{i_2}]$ . The fused  $J$ -dependent weights one has to go through take the following  $q$ -hypergeometric form [Man14], [Bor17], [CP16]:

$$w_{x,s}^{(J)}(i_1, j_1; i_2, j_2) = \mathbf{1}_{i_1+j_1=i_2+j_2} \frac{(-1)^{i_1+j_2} t^{\frac{1}{2}i_1(i_1+2j_1-1)} s^{j_2-i_1} x^{i_1} (t; t)_{j_1} (s^2; t)_{i_2} (xs^{-1}; t)_{j_1-i_2}}{(t; t)_{j_2} (t; t)_{i_2} (s^2; t)_{i_1} (xs; t)_{i_1+j_1}} \times {}_4\bar{\phi}_3 \left( \begin{matrix} t^{-i_1}; t^{-i_2}, t^J sx, t sx^{-1} \\ s^2, t^{1+j_1-i_2}, t^{1+J-i_1-j_1} \end{matrix} \middle| t, t \right).$$

Here  ${}_{r+1}\bar{\phi}_r$  is the regularized terminating basic hypergeometric series:

$$\begin{aligned} {}_{r+1}\bar{\phi}_r \left( \begin{matrix} t^{-n}; a_1, \dots, a_r \\ b_1, \dots, b_r \end{matrix} \middle| t, z \right) &= \sum_{k=0}^n z^k \frac{(t^{-n}; t)_k}{(t; t)_k} \prod_{i=1}^r (a_i; t)_k (b_i t^k; t)_{n-k} \\ &= \prod_{i=1}^r (b_i; t)_n \cdot {}_{r+1}\phi_r \left( \begin{matrix} t^{-n}, a_1, \dots, a_r \\ b_1, \dots, b_r \end{matrix} \middle| t, z \right). \end{aligned} \quad (6.12)$$

The displayed spin-fused weight  $w_{x,s}^{(J)}$  reduces to the spin Hall–Littlewood weight of Remark 6.2 at  $J = 1$ , and to (6.10) in the limit  $s \rightarrow 0$ . The latter is a removable singularity and not a substitution: already at  $J = 1$  and  $(i_1, j_1; i_2, j_2) = (1, 0; 1, 0)$  the factors  $s^{j_2-i_1}$  and  $(xs^{-1}; t)_{j_1-i_2}$  blow up and vanish separately, while the whole expression equals  $(1 - stx)/(1 - sx)$  and tends to the spinless value 1. Here  $s$  is the parameter of the vertical (generic Verma) representation of  $U_t(\widehat{\mathfrak{sl}}_2)$ . Setting  $s^2 = t^{-I}$  caps the vertical occupations at  $I$ .

## 7 Deformed weights and particle systems

### 7.1 The $q$ -Whittaker case and $q$ -PushTASEP

Replace the Schur weights of Section 3 by their  $q$ -Whittaker deformation: the up-right weight  $W$  (6.2) and the corresponding down-right dual  $W^*$  (6.3). The YBE (2.9) persists, with a  $q$ -deformed cross  $\mathbb{R}_{x,y}^{(q)}$  in place of (2.8). The cross is the unique (up to scale) solution of (2.9); it obeys the same arrow conservation  $i_1 + j_2 = j_1 + i_2$  as (2.8), depends on  $x, y$  only through the product  $xy$ , and degenerates to (2.8) as  $q \rightarrow 0$ . Its entries include

$$\begin{aligned} \mathbb{R}_{x,y}^{(q)}(0, 0; m, m) &= \frac{(xy)^m}{(q; q)_m}, & \mathbb{R}_{x,y}^{(q)}(m, m; 0, 0) &= (q; q)_m, \\ \mathbb{R}_{x,y}^{(q)}(1, 1; 1, 1) &= xy + q, & \mathbb{R}_{x,y}^{(q)}(1, 2; 1, 2) &= \mathbb{R}_{x,y}^{(q)}(2, 1; 2, 1) = xy(1 + q) + q^2, \\ \mathbb{R}_{x,y}^{(q)}(2, 2; 1, 1) &= (1 - q^2)(xy + q + q^2), & \mathbb{R}_{x,y}^{(q)}(2, 2; 2, 2) &= (xy)^2 + xyq(1 + q)^2 + q^4, \end{aligned} \quad (7.1)$$

and in general it is the regularized terminating  ${}_2\bar{\phi}_1$  series

$$\mathbb{R}_{x,y}^{(q)}(i_1, j_1; i_2, j_2) = \mathbf{1}_{i_1+j_2=j_1+i_2} \frac{(q; q)_{j_1}}{(q; q)_{j_2} (q; q)_{i_2}} (xy)^{i_2} {}_2\bar{\phi}_1 \left( \begin{matrix} q^{-i_2}; q^{-i_1} \\ q^{1+j_2-i_2} \end{matrix} \middle| q, \frac{q^{i_1+j_2+1}}{xy} \right), \quad (7.2)$$

where  ${}_2\bar{\phi}_1$  is the regularized terminating series (6.12). Written out, the  $q$ -deformed cross is the finite sum

$$\begin{aligned} \mathbb{R}_{x,y}^{(q)}(i_1, j_1; i_2, j_2) &= \mathbf{1}_{i_1+j_2=j_1+i_2} \frac{(q; q)_{j_1}}{(q; q)_{j_2} (q; q)_{i_2}} \\ &\times \sum_{k=0}^{i_2} \frac{(q^{-i_2}; q)_k (q^{-i_1}; q)_k}{(q; q)_k} (q^{1+j_2-i_2+k}; q)_{i_2-k} q^{(i_1+j_2+1)k} (xy)^{i_2-k}. \end{aligned} \quad (7.3)$$

**Remark 7.1.** This  $q$ -deformation of the benign expression  $(xy)^{\min(i_2, j_2)}$  in (2.8) may appear contrived, but it is required to satisfy the YBE. There also exists a further spin deformation, see [BMP21], [MP22].

**Remark 7.2.** The sum over the outputs of the cross is independent of  $(i_1, j_1)$ :

$$\sum_{i_2, j_2} \mathbb{R}_{x,y}^{(q)}(i_1, j_1; i_2, j_2) = \frac{1}{(xy; q)_\infty} \quad \text{for every } (i_1, j_1). \quad (7.4)$$

For  $(i_1, j_1) = (0, 0)$  this is the (infinite)  $q$ -binomial theorem

$$\sum_{m \geq 0} \frac{(xy)^m}{(q; q)_m} = \frac{1}{(xy; q)_\infty}.$$

This identity is the  $q$ -Whittaker Cauchy kernel. For  $q = 0$ , it reduces to the geometric series  $\sum_{a \geq 0} (xy)^a = 1/(1 - xy)$  which appeared throughout the Schur case.

Let us apply the bijectivization in the same setting as the one which produced the RSK correspondence in Section 3.2.

In the bulk (we assume  $0 < q < 1$  and  $x, y > 0$  here; the summands are then nonnegative, and zero-weight states are discarded as in Section 5) the  $q$ -deformation turns the deterministic move of Lemma 3.2 into a random coupling:  $k_1, k'_1$  still range over the equal-length intervals (3.5), but the summands (3.4) now carry  $q$ -binomial factors, so, for all but special boundary occupations,<sup>8</sup> only the two *sums* (and not the individual summands) match, and so  $\mathbf{p}^{\text{fwd}}(k_1 \rightarrow k'_1)$  must spread the probabilistic mass.

For the instance of Figure 3, with boundary occupations  $(i_1, i_2, i_3; j_1, j_2, j_3) = (3, 2, 4; 3, 4, 2)$ , each of the six summands (3.4) deforms into  $x^4 y^3 (1 + q)(1 + q^2)$  times a polynomial in  $xy$  and  $q$ . Dividing out this common factor, the two sides of (2.9) read

$$\begin{aligned} &\underbrace{(1 + q^2 + q^4)(1 + q + q^2 + q^3 + q^4)}_{k_1=1} + \underbrace{(q^2 + xy)(1 + q + q^2)(1 + q + q^2 + q^3 + q^4)}_{k_1=2} \\ &\quad + \underbrace{(xy)^2(1 + q + q^2) + q^2 xy(1 + q)(1 + q + q^2) + q^6}_{k_1=3} \\ &= \underbrace{(1 + q^2)(1 + q + q^2 + q^3 + q^4)}_{k'_1=0} + \underbrace{(1 + q^2)(1 + q + q^2)(q^2(1 + q + q^2) + xy(1 + q))}_{k'_1=1} \\ &\quad + \underbrace{(1 + q + q^2)((xy)^2 + q^2 xy(1 + q + q^2) + q^6)}_{k'_1=2}, \end{aligned} \quad (7.5)$$

<sup>8</sup>For example, at  $(i_1, i_2, i_3; j_1, j_2, j_3) = (0, 0, 1; 0, 0, 1)$  both lists are  $\{1, xy\}$  and the matching stays deterministic.

where the left summands are indexed by  $k_1$  as in (3.5), with  $\varkappa_1 = 7 - k_1$  and  $\nu_2 = 4 + k'_1$  in the notation of Section 1.2. At  $q = 0$ , the display collapses to  $1 + xy + (xy)^2$  on both sides, which is (1.2) stripped of  $x^4y^3$ . For  $0 < q < 1$  no term-by-term matching survives: the  $k_1 = 1$  and  $k'_1 = 0$  summands are the only ones free of  $xy$ , and they are different polynomials in  $q$ . At  $q = \frac{1}{2}$  and  $x = y = \frac{1}{2}$  the six summands become proportional to  $651 + 434 + 74 = 620 + 455 + 84$ , with no value shared between the two sides, and the independent coupling — with  $\mathbf{p}^{\text{fwd}}(k_1 \rightarrow k'_1)$  proportional to the  $k'_1$ -th summand, for every  $k_1$  — has  $\mathbf{p}^{\text{fwd}} = \frac{620}{1159}, \frac{455}{1159}, \frac{84}{1159}$  for  $k'_1 = 0, 1, 2$ , respectively. (At  $xy = 1$  the two sides accidentally coincide as multisets,  $651 + 1085 + 620 = 620 + 1085 + 651$ ; an equality of values at a single point does not produce a matching of the polynomial summands.)

However, at the leftmost column the bijectivization becomes unique after summing the left-hand side of (2.9) over the internal occupations. There the cross pushes its arrows into the system through the reservoir of infinitely many vertical paths, see Remark 2.11 and Figure 15. The vertex weights with infinitely many vertical arrows do not depend on the incoming horizontal occupation:  $W(\infty, j_1; \infty, j_2) = x^{j_2}/(q; q)_{j_2}$ , the  $q$ -deformation of Remark 2.11. Hence, in the YBE with  $i_1, i_2$  fixed, we may sum over the internal occupations  $k_1, k_2$  in the left-hand side of (2.9), and by (7.4) that side collapses to a *single* term:

$$\frac{1}{(xy; q)_\infty} \frac{x^{j_2}}{(q; q)_{j_2}} \frac{y^{j_1}}{(q; q)_{j_1}} = \sum_{k'_1, k'_2} \frac{x^{k'_2}}{(q; q)_{k'_2}} \frac{y^{k'_1}}{(q; q)_{k'_1}} \mathbb{R}_{x,y}^{(q)}(k'_2, k'_1; j_2, j_1). \quad (7.6)$$

Note that both sides are independent of the fixed  $i_1, i_2$ . A bijectivization of an identity with a singleton on one side is unique (Section 5). Under it, the pair  $(k'_1, k'_2)$  is drawn with probability proportional to its weight in the right-hand side of (7.6).

Let us use the symmetry

$$\frac{x^{k'_2} y^{k'_1}}{(q; q)_{k'_2} (q; q)_{k'_1}} \mathbb{R}_{x,y}^{(q)}(k'_2, k'_1; j_2, j_1) = \frac{x^{j_2} y^{j_1}}{(q; q)_{j_2} (q; q)_{j_1}} \mathbb{R}_{x,y}^{(q)}(j_2, j_1; k'_2, k'_1)$$

of the  $q$ -deformed cross, and denote these probabilities to pick  $(k'_1, k'_2)$  by

$$\mathbb{L}_{x,y}(i_1, j_1; i_2, j_2) := \frac{\mathbb{R}_{x,y}^{(q)}(i_1, j_1; i_2, j_2)}{\sum_{i'_2, j'_2} \mathbb{R}_{x,y}^{(q)}(i_1, j_1; i'_2, j'_2)} = (xy; q)_\infty \mathbb{R}_{x,y}^{(q)}(i_1, j_1; i_2, j_2). \quad (7.7)$$

In the regime  $0 \leq q < 1$ ,  $x, y \geq 0$  with  $xy < 1$  of Section 2.1, we have  $\mathbb{L}_{x,y} \geq 0$  and  $\sum_{i_2, j_2} \mathbb{L}_{x,y}(i_1, j_1; i_2, j_2) = 1$ . Both facts are established in the Appendix of [BMP21] for the more general spin weights: the constant row sum is [BMP21, Proposition A.5], and nonnegativity is [BMP21, Proposition A.8], proved there for  $q \in (0, 1)$  and spin parameter  $s \in [-\sqrt{q}, 0)$ ; the  $s = 0$  weights (7.3) are covered by the limit  $s \rightarrow 0^-$ . Proposition A.8 also assumes its spectral parameters lie in  $[-s, -s^{-1}]$ ; for fixed  $x, y > 0$  this holds for all  $s < 0$  close enough to 0. The boundary cases  $q = 0$ ,  $x = 0$ , or  $y = 0$  follow by continuity, or directly from the  $q = 0$  Schur weights.

Let us explain a connection to particle systems. Applied across the lattice, the moves grow a field  $\{\lambda^{(m,n)}\}$  of Young diagrams on the quadrant  $\mathbb{Z}_{\geq 0}^2$  — a  $q$ -Whittaker Yang–Baxter field — whose *first parts*  $\lambda_1^{(m,n)}$  encode the particles of the (discrete-time) *geometric  $q$ -PushTASEP* of [MP17, §6.3] (the  $s = 0$  case of the  $q$ -Hahn PushTASEP of [CMP21, §3.2.1]). Concretely: set  $\lambda^{(m,0)} = \lambda^{(0,n)} = \emptyset$  on the axes, fix an admissible forward kernel for every bulk instance once and for all, and apply these kernels cell by cell in increasing  $m + n$ , following [BMP21]. The bulk coupling is not unique — any bijectivization of the bulk instances of (2.9) may be used; the resulting field depends on this choice, while the marginals used below do not [BP19], [BMP21]. Along any down-right cut

(a lattice path in  $\mathbb{Z}_{\geq 0}^2$  made of unit steps to the right and down) the field has the law of the  $q$ -Whittaker process (the  $t = 0$  Macdonald process) of [BC14] — whence the name — independently of the bulk coupling. The four diagrams at the corners of a unit cell,

$$\varkappa = \lambda^{(m,n)}, \quad \mu = \lambda^{(m+1,n)}, \quad \lambda = \lambda^{(m,n+1)}, \quad \nu = \lambda^{(m+1,n+1)},$$

obey the local rule which randomly picks  $\nu$  based on  $\mu, \varkappa, \lambda$  and an integer  $k$  (the analogue of the RSK input  $k_{ab}$  from the proof of Proposition 3.5). The input  $k$  is now random:  $q$ -geometric with parameter  $x_{n+1}y_{m+1}$ , that is,  $\mathbb{P}(k) = (x_{n+1}y_{m+1}; q)_\infty (x_{n+1}y_{m+1})^k / (q; q)_k$ , independently over the cells [MP17, §6.2]. The kernel  $\mathbb{L}$  enforces the mandatory lower bound  $\nu_1 - \lambda_1 \geq \max(0, \mu_1 - \lambda_1)$ , but the excess over it is not in general the independent input  $k$ : later particles also receive a random push. The jump is bare  $q$ -geometric, as in (7.10), whenever the preceding particle does not move, which is automatic for the leading particle. At  $q = 0$  the inputs reduce to the geometric matrix entries feeding RSK. Thanks to the unique bijectivization at the zeroth column, the evolution of the first parts of the partitions is *marginally Markovian*, governed by the stochasticized cross  $\mathbb{L}_{x_{n+1}, y_{m+1}}(i_1, j_1; i_2, j_2)$  with

$$(i_1, j_1) = (\lambda_1 - \varkappa_1, \mu_1 - \varkappa_1), \quad (i_2, j_2) = (\nu_1 - \mu_1, \nu_1 - \lambda_1), \quad i_1 + j_2 = j_1 + i_2 = \nu_1 - \varkappa_1. \quad (7.8)$$

The first coordinate of the field plays the role of time. The update (7.8) involves only the first parts, so the array  $(\lambda_1^{(m,n)})_{n \geq 1}$  evolves as a Markov chain in  $m$ , which we describe as a particle system. Set

$$P_n(m) := \lambda_1^{(m,n)} + n, \quad n \geq 1, \quad P_0 \equiv 0. \quad (7.9)$$

By the interlacing,  $P_0 < P_1(m) < P_2(m) < \dots$ , and the particles move to the right, pushing one another. One step  $m \rightarrow m+1$  updates them in order  $n = 1, 2, \dots$ : given the move  $\ell$  of  $P_{n-1}$  and the gap  $g = P_n(m) - P_{n-1}(m) - 1$ , the particle  $P_n$  moves right by  $L$  with probability  $\mathbb{L}_{x_n, y_{m+1}}(g, \ell; g + L - \ell, L)$ . Indeed, the four counts inside  $\mathbb{L}_{x_n, y_{m+1}}$  in the latter expression can be matched to (7.8) for the cell with the lower left corner  $\varkappa = \lambda^{(m, n-1)}$ , with the outgoing pair  $(g + L - \ell, L)$  recording the new gap and the jump. In particular, the leading particle  $P_1$  receives “push”  $\ell = 0$  from the frozen  $P_0$  and jumps  $q$ -geometrically,

$$\mathbb{L}_{x_1, y_{m+1}}(g, 0; g + L, L) = (x_1 y_{m+1}; q)_\infty \frac{(x_1 y_{m+1})^L}{(q; q)_L}, \quad (7.10)$$

independently of the gap  $g$ . The dynamics preserves the order:  $\mathbb{L}_{x_n, y_{m+1}} = 0$  unless  $L \geq \ell - g$ , so  $\ell > g$  forces a push.

A continuous-time limit of the  $q$ -PushTASEP (set all  $y_m = \varepsilon$  and observe the chain after  $\lfloor \tau/\varepsilon \rfloor$  steps at continuous time  $\tau$ , as  $\varepsilon \rightarrow 0$ ) produces a simpler dynamics introduced in [BP16] and further studied in [CP15]. Under the continuous-time  $q$ -PushTASEP, each particle  $n$  carries an independent exponential clock with rate  $x_n/(1-q)$ ; when it rings the particle steps forward by one and, with probability  $q^g$  (with  $g$  the gap to the next particle ahead before the step), instantaneously pushes that neighbor forward by one as well, the push cascading down the line. At  $q = 0$  the push probability is 1 across a zero gap and 0 otherwise, so a jump shoves the whole block of adjacent particles ahead by one. That is, for  $q = 0$  we get back the classical PushTASEP.

## 7.2 Hall–Littlewood: a row-indexed RSK and the six-vertex model

The Hall–Littlewood weights (6.1) — the  $t$ -deformation of the higher-spin model of Remark 2.4 — carry a randomized bijectivization parallel to the  $q$ -Whittaker one of Section 7.1. The *horizontal*

edges in the Hall–Littlewood case have capacity one ( $j_1, j_2 \in \{0, 1\}$ ), while the vertical edges stay unbounded ( $i_1, i_2 \in \mathbb{Z}_{\geq 0}$ ) and record multiplicities of the *row* lengths of the partitions, rather than of their column lengths as in Section 7.1 (cf. Remark 2.10). In this sense the insertion is indexed by rows, and, as we now explain, its scalar marginal is the *stochastic six-vertex model* rather than the  $q$ -PushTASEP.

Recall the vertex weights entering the Hall–Littlewood Cauchy identity (6.4). The up-right weight is (6.1), in formulas

$$W(i_1, j_1; i_2, j_2) = \mathbf{1}_{i_1+j_1=i_2+j_2} \times \begin{cases} 1, & (j_1, j_2) = (0, 0), \\ x, & (j_1, j_2) = (1, 1), \\ 1, & (j_1, j_2) = (1, 0), \\ x(1 - t^{i_1}), & (j_1, j_2) = (0, 1), \end{cases} \quad (7.11)$$

whose  $N$ -row partition function is  $P_\lambda(x_1, \dots, x_N; 0, t)$  (Proposition 6.1). The  $y$ -side of (6.4) carries the functions  $Q_\lambda$ , whose branching rule differs from that of  $P_\lambda$ : the factor  $1 - t^m$  is attached to a path *entering* a column whose multiplicity becomes  $m$ , rather than leaving a column of multiplicity  $m$  [Mac95, Ch. III, §5]. Accordingly, the down-right weight (the analogue of the reflection (2.7), with the conservation law  $i_2 + j_1 = i_1 + j_2$ ) is

$$W^*(i_1, j_1; i_2, j_2) = \mathbf{1}_{i_2+j_1=i_1+j_2} \times \begin{cases} 1, & (j_1, j_2) = (0, 0), \\ y, & (j_1, j_2) = (1, 1), \\ 1 - t^{i_1}, & (j_1, j_2) = (1, 0), \\ y, & (j_1, j_2) = (0, 1). \end{cases} \quad (7.12)$$

Gluing  $N$  rows of the up-right model and  $M$  rows of the down-right model along a common boundary  $\lambda$ , as in Figure 13, produces the vertex model for the left-hand side of the Cauchy identity (6.4).

The YBE (2.9) with the weights (7.11)–(7.12) has the unique (up to scale) solution

$$\begin{aligned} \mathbb{R}_{x,y}^{(t)}(0, 0; 0, 0) &= 1, & \mathbb{R}_{x,y}^{(t)}(1, 0; 1, 0) &= \mathbb{R}_{x,y}^{(t)}(0, 1; 0, 1) = \frac{1 - txy}{1 - xy}, & \mathbb{R}_{x,y}^{(t)}(1, 1; 1, 1) &= t, \\ \mathbb{R}_{x,y}^{(t)}(0, 0; 1, 1) &= \frac{xy(1 - t)}{1 - xy}, & \mathbb{R}_{x,y}^{(t)}(1, 1; 0, 0) &= \frac{1 - t}{1 - xy}. \end{aligned} \quad (7.13)$$

Like the  $q$ -Whittaker cross (7.2), it depends on  $x, y$  only through the product  $xy$ .

**Remark 7.3.** All row sums of the cross coincide (cf. Remark 7.2):

$$\sum_{i_2, j_2 \in \{0, 1\}} \mathbb{R}_{x,y}^{(t)}(i_1, j_1; i_2, j_2) = \frac{1 - txy}{1 - xy} \quad \text{for every } (i_1, j_1), \quad (7.14)$$

which is the building block for the Hall–Littlewood Cauchy kernel.

Dragging the  $NM$  crosses  $\mathbb{R}_{x_i, y_j}^{(t)}$ , one for each of the  $NM$  intersections of the  $N$  up-right lines with the  $M$  down-right ones, through the lattice exactly as in Section 2.5 yields the Yang–Baxter proof of the Cauchy identity (6.4). This Yang–Baxter proof is essentially the same as the one in [WZJ16], [Bor17], and can be traced back to the  $q$ -boson realization of the Hall–Littlewood functions in [Tsi06].

Stochasticize the cross as in (7.7),

$$\mathbb{L}_{x,y}^{(t)}(i_1, j_1; i_2, j_2) := \frac{1-xy}{1-txy} \mathbb{R}_{x,y}^{(t)}(i_1, j_1; i_2, j_2), \quad (7.15)$$

and abbreviate

$$b_1 := \frac{1-xy}{1-txy}, \quad b_2 := t \frac{1-xy}{1-txy} = t b_1, \quad 0 \leq b_2 \leq b_1 \leq 1. \quad (7.16)$$

The six transition probabilities become

$$\begin{aligned} \mathbb{L}_{x,y}^{(t)}(1, 0; 1, 0) &= \mathbb{L}_{x,y}^{(t)}(0, 1; 0, 1) = 1, \\ \mathbb{L}_{x,y}^{(t)}(0, 0; 0, 0) &= b_1, \quad \mathbb{L}_{x,y}^{(t)}(0, 0; 1, 1) = 1 - b_1, \\ \mathbb{L}_{x,y}^{(t)}(1, 1; 1, 1) &= b_2, \quad \mathbb{L}_{x,y}^{(t)}(1, 1; 0, 0) = 1 - b_2. \end{aligned} \quad (7.17)$$

Figure 27, top row.

As in the  $q$ -Whittaker case, at the leftmost column the bijectivization is unique. With the reservoir weights  $W(\infty, j_1; \infty, j_2) = x^{j_2}$  and  $W^*(\infty, j_1; \infty, j_2) = y^{j_2}$  (Remark 2.11; the factors  $1 - t^\infty = 1$  disappear), summing the YBE over the internal occupations collapses its left-hand side to a single term, and the analogue of (7.6) reads

$$\frac{1-txy}{1-xy} x^{j_2} y^{j_1} = \sum_{k'_1, k'_2 \in \{0,1\}} x^{k'_2} y^{k'_1} \mathbb{R}_{x,y}^{(t)}(k'_2, k'_1; j_2, j_1). \quad (7.18)$$

The same symmetry of the cross as in the  $q$ -Whittaker case identifies the resulting unique coupling with  $\mathbb{L}_{x,y}^{(t)}$  (7.15).

Dragging the crosses through the Hall–Littlewood Cauchy lattice — the construction of [BP19] — grows a field  $\{\lambda^{(m,n)}\}$  of partitions on the quadrant, with  $\lambda^{(m,n)}$  distributed as the Hall–Littlewood measure

$$\text{Prob}(\lambda) \propto P_\lambda(x_1, \dots, x_n; 0, t) Q_\lambda(y_1, \dots, y_m; 0, t), \quad (7.19)$$

the  $t$ -analogue of the Schur measure. Along any down-right cut the field is the *Hall–Littlewood process*, the  $t$ -deformation of the Schur process [OR03]. In the bulk the coupling is randomized (as in Section 7.1), while at the leftmost column it is the unique (7.18). Around a unit cell with corners  $\varkappa = \lambda^{(m,n)}$ ,  $\mu = \lambda^{(m+1,n)}$ ,  $\lambda = \lambda^{(m,n+1)}$ ,  $\nu = \lambda^{(m+1,n+1)}$ , the stochasticized cross  $\mathbb{L}_{x_{n+1}, y_{m+1}}^{(t)}$  governs the *lengths* of the partitions:

$$(i_1, j_1) = (\ell(\lambda) - \ell(\varkappa), \ell(\mu) - \ell(\varkappa)), \quad (i_2, j_2) = (\ell(\nu) - \ell(\mu), \ell(\nu) - \ell(\lambda)), \quad (7.20)$$

with  $i_1 + j_2 = j_1 + i_2 = \ell(\nu) - \ell(\varkappa)$ . This is the row-indexed counterpart of (7.8), with the first parts  $\lambda_1$  replaced by the lengths  $\ell(\lambda) = \lambda'_1$  (cf. Remark 2.10). In particular, the array of lengths — equivalently, of the numbers of zero parts  $m_0(\lambda^{(m,n)}) = n - \ell(\lambda^{(m,n)})$  — evolves in a marginally Markovian way.

To recognize this marginal, recall the stochastic six-vertex model [GS92], [BCG16]. The general six-vertex model on  $\mathbb{Z}^2$  has six admissible arrow configurations around a vertex, weighted  $a_1, a_2, b_1, b_2, c_1, c_2 \geq 0$  (see Figure 27, bottom row, and [Bax07]). The weights are called *stochastic* [GS92] if

$$a_1 = a_2 = 1, \quad b_1 + c_1 = b_2 + c_2 = 1, \quad b_1, b_2 \in [0, 1]. \quad (7.21)$$

Then the weight of each vertex is the conditional probability of its outgoing arrows (through the top and the right) given the incoming ones (from below and from the left), cf. Figure 25, and the model in the quadrant  $\mathbb{Z}_{\geq 1}^2$  is sampled by a Markov chain, vertex by vertex along the successive diagonals  $\{m + n = \text{const}\}$ . Impose the *infinite domain wall* boundary conditions: a path enters from the left at every row, and no paths enter through the bottom. The *height function* is

$$H(m, n) := \#\{\text{up-right paths passing strictly to the right of } (m, n)\}, \quad 0 \leq H(m, n) \leq n, \quad (7.22)$$

see Figure 28 for an illustration.

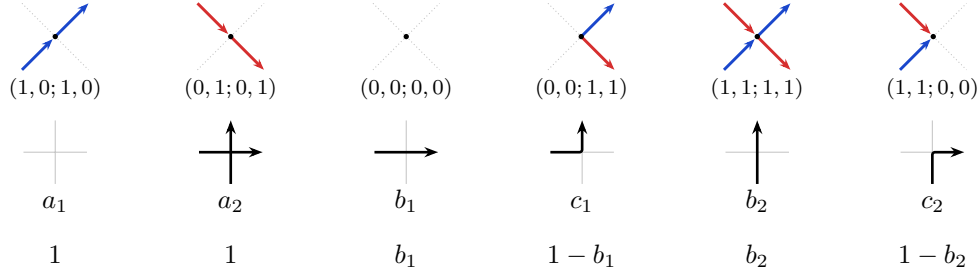


Figure 27: Top row: the six states  $(i_1, j_1; i_2, j_2)$  of the stochasticized cross  $\mathbb{L}_{x,y}^{(t)}$  (blue = the  $P$ -line carrying  $x$ , red = the  $Q$ -line carrying  $y$ ; dotted = empty). Bottom row: the corresponding vertices of the stochastic six-vertex model, in the standard order  $a_1, a_2, b_1, c_1, b_2, c_2$ . The cross state  $(i_1, j_1; i_2, j_2)$  becomes the vertex whose (bottom, left; top, right) occupations are  $(j_1, 1-i_1; j_2, 1-i_2)$ .

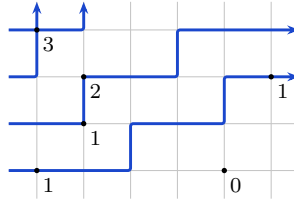


Figure 28: A configuration of the stochastic six-vertex model in the quadrant with the infinite domain wall boundary conditions, and several values of the height function (7.22). Paths leaving the picture on the right continue and eventually turn up.

The next statement is proven in the same way as its  $q$ -Whittaker counterpart from Section 7.1:

**Proposition 7.4** ([Bor18], [BBW16], [BM18], [BP19]). *Under the Yang–Baxter field  $\{\lambda^{(m,n)}\}$  constructed from the bijectivization of the Hall–Littlewood YBE, the array  $m_0(\lambda^{(m,n)})$  evolves as the height function of the stochastic six-vertex model in the quadrant with the infinite domain wall boundary conditions, with  $b_1, b_2$  given by (7.16) under  $xy \rightsquigarrow x_n y_m$  at the coordinate  $(m, n)$ .*

With homogeneous parameters  $x_n y_m \equiv xy$ , in the Poisson-type limit around the diagonal ( $xy \rightarrow 1$ , so that  $b_1, b_2 \rightarrow 0$  with the ratio  $b_2/b_1 = t$  fixed by (7.16)) the six-vertex model degenerates to the asymmetric simple exclusion process (ASEP) with left and right jump rates  $t$  and 1 [BCG16]. Parameter convergence alone is not enough here: one observes the particle configuration along the diagonal and accelerates the diagonal coordinate by the order  $b_1^{-1}$ , so that jumps, which occur with probabilities of order  $b_1$  and  $b_2$ , happen at rates 1 and  $t$  per unit of limiting time. See [BCG16] for the precise scaling, and [Agg17] for the proof and the mode of convergence. This is the row-indexed counterpart of the  $q$ -PushTASEP produced from columns in Section 7.1. At  $t = 0$  the

Hall–Littlewood weights (6.1) become the Schur weights (Remark 2.4), and the height field reduces to the corresponding Schur-measure observable. The particle system along the diagonal becomes the continuous-time TASEP.

## 8 Further directions

Reading the Yang–Baxter equation as a *probabilistic normalization* identity between two random configurations, and not only an algebraic one, is fruitful: dragging a cross through a lattice gave the Cauchy identity (Section 2.5), RSK and its deformations (Sections 3 and 7), and their particle-system marginals. Here we collect further directions in which this reading has been used.

### 8.1 Stationary measures via the forgetting property

For the stochastic vertex models of these notes, sending the number of arrows carried by one line to infinity turns that line into an inexhaustible reservoir that “forgets” how many arrows entered, so its output is independent of its input — a property of the specific weights, not of stochasticity alone (we saw it first in Remark 2.11). Then, the YBE allows one to exchange this infinite line with another, finite stochastic line (encoding the evolution of a particle system). Since the distribution before and after applying the stochastic line is the same, such “reservoir measures” are *stationary* for particle systems. This approach was used to construct stationary measures for the colored ASEP, the colored  $q$ -Boson process, and the colored  $q$ -PushTASEP, on the ring and on the line (the latter through a stationary vertex model in the quadrant) by Aggarwal–Nicoletti–Petrov [ANP25]. For a single color the quadrant construction is due to Aggarwal [Agg18].

The reservoir weights are a lattice form of the multiline queues of Ferrari–Martin [FM07] and Martin [Mar20], and the underlying single-color Burke property appeared earlier for directed polymers in O’Connell–Yor [OY01] and Seppäläinen [Sep12]. Stationary measures for multiclass particle systems also admit an algebraic description via the Matrix Product Ansatz, which goes back to Derrida–Evans–Hakim–Pasquier [DEHP93] and Derrida–Janowsky–Lebowitz–Speer [DJLS93], and was extended to several species by Mallick–Mallick–Rajewsky [MMR99] and Prolhac–Evans–Mallick [PEM09]; see the survey of Blythe–Evans [BE07]. The queueing constructions of Angel [Ang06] and Ferrari–Martin [FM06], [FM09] give a combinatorial counterpart, and letting the number of classes grow leads to the TASEP speed process of Amir–Angel–Valkó [AAV11]. The stationary measures tie to nonsymmetric Macdonald polynomials, as in Cantini–de Gier–Wheeler [CdGW15], Corteel–Mandelstam–Williams [CMW22], and Borodin–Wheeler [BW22b], [BW22a]; and, for the colored  $q$ -Boson process, to modified Macdonald polynomials, as in Ayer–Mandelstam–Martin [AMM23], [AMM24].

### 8.2 Symmetry in the spectral parameters

The partition functions of Sections 2 and 6 build symmetric polynomials one row at a time, feeding in the spectral parameters in a fixed order; that they come out symmetric is a corollary of the YBE, since an  $R$ -cross (of a different form than the ones we use here) dragged through two adjacent rows interchanges their parameters while fixing the partition function. Bijectivizing this single swap turns the symmetry into a local Markov move on semistandard Young tableaux of a fixed shape: the move preserves the partition function but carries the tableau distribution for one order of the parameters to the one for the swapped order. It is a probabilistic version of the Bender–Knuth involution [BK72].

Iterating this single swap up an infinite interlacing array drags the bottom parameter off to infinity, and homogeneity turns the residual rescaling into a shift  $t \mapsto qt$  of the process time (here  $q$  is the ratio of the geometric family of TASEP speeds and  $t$  is the running time — not the Macdonald parameters of Section 6). In the limit  $q = e^{-\varepsilon} \rightarrow 1$  with  $\lfloor \tau/\varepsilon \rfloor$  iterations of this drag, this becomes a time-homogeneous *backward* process: Petrov–Saenz [PS22] build a backward Hammersley-type process with  $\mu_t \mathbf{L}_\tau = \mu_{e^{-\tau}t}$ , mapping the fixed-time law  $\mu_t$  of TASEP with the step initial condition to the law at an earlier time  $e^{-\tau}t$ .

The  $q$ -Hahn TASEP goes back to Povolotsky [Pov13] and Corwin [Cor15]; in its multiparameter version, due to Borodin–Petrov [BP18, §6.6], each particle  $x_n$  (a position, not a spectral parameter) carries its own jump parameter  $\nu_n$  (coming from the vertex model spin parameter). The parameter symmetry described next is from Petrov [Pet21]. For  $0 < \nu_{n+1} < \nu_n < 1$  the swap  $\nu_n \leftrightarrow \nu_{n+1}$  is a  $q$ -deformed beta-binomial redistribution of  $x_n$ , reproducing the process with the two parameters swapped at every fixed time. The swap exists because, from the step initial configuration  $x_n(0) = -n$ , transposing  $\nu_n$  and  $\nu_{n+1}$  leaves the joint law of all the other particles unchanged. Both statements need that initial configuration; for other deterministic data the symmetry typically fails already after a single update. The same elementary swap degenerates to  $q$ -TASEP and to the directed beta polymer, where it becomes a Beta-splitting of two adjacent partition functions. A random matrix shadow of this probabilistic symmetry was explored in Petrov–Tikhonov [PT20].

The symmetry lifts from fixed-time marginals to whole trajectories. Indeed, Petrov–Saenz [PS24] constructed “rewriting history” Markov operators on the space of trajectories of a particle system such as TASEP. These operators act by resampling a particle’s entire past trajectory while carrying the joint law onto that of the parameter-swapped process. Dragging to infinity gives the shift intertwiner, whose continuous limit is a Lax equation  $t \dot{\mathbf{T}} = [\mathbf{B}, \mathbf{T}]$  for the TASEP semigroup  $\mathbf{T} = \mathbf{T}(t)$ , with  $\mathbf{B}$  the generator of the backward process. This last step is formal: [PS24] exhibits no common invariant domain on which  $\mathbf{B}\mathbf{T}$  and  $\mathbf{T}\mathbf{B}$  are both defined. The Lax equation has the same form for  $q$ -TASEP and TASEP, and differs only in the form of the generator  $\mathbf{B}$  of the backward process.

### 8.3 Shift-invariance and hidden symmetries

The transposition of spectral parameters also surfaces in identities between *distant* observables, though the phenomenon is less the swap itself than a hidden symmetry that the swap merely records. Borodin–Gorin–Wheeler [BGW22] prove (under certain admissibility conditions) that the joint law of colored height functions of the inhomogeneous colored stochastic six-vertex model is invariant under moving one observation point up by one lattice step and raising its color cutoff by one, at the cost of transposing two row spectral parameters. The identity is nonlocal, as the swapped rows may be arbitrarily far apart. The proof is analytic, by Lagrange interpolation in the spectral parameters at the nodes where a row parameter equals a column parameter and the  $R$ -matrix degenerates to a permutation. Fusion, analytic continuation in the spin, and scaling limits push the invariance down to the Beta, Gamma, and O’Connell–Yor polymers, the KPZ equation, and the Airy sheet. The degenerate specialization of the vertex model identity is the color–position symmetry of Borodin–Wheeler [BW22a] and Borodin–Bufetov [BB21].

Independently, Dauvergne [Dau22] settled a conjecture of [BGW22] for last passage percolation and directed polymers, by a route close to the present work: the RSK and geometric RSK correspondences together with a decoupling property. The cleanest formulation, due to Galashin [Gal21], is a single *flip* invariance of the partition function (a 180° rotation of the left-to-right boundary connections, with the bottom-to-top ones fixed and the row spectral parameters reversed), of which

the shift is a composition of two flips; the proof is algebraic, through the Yang–Baxter basis of the Hecke algebra. See also Bufetov [Buf20] and Zhang [Zha23]. The two sides of a flip carry *equal numbers* of configurations, yet the weights match only in aggregate. It would be interesting to find useful couplings of the two sides, as we have done for the Yang–Baxter equation and Cauchy identities in this work.

## 8.4 Markov duality

Markov duality extends the integrable structure to the *dynamics* of observables of particle systems coming from stochastic vertex models. We use the quantum-group vocabulary below — co-product, universal  $R$ -matrix — only as pointers to the literature, without definitions. The symmetry at work is now the co-product rather than the spectral parameter: the stochastic Markov generator of a particle system (for discrete-time chains, the transition operator) commutes with the  $U_q$ -action; combined with a reversible measure or an explicit conjugation, this symmetry produces a *duality function*  $D(\eta, \xi)$  intertwining the generator with the transpose of a dual generator,

$$\mathbb{E}_\eta[D(\eta_t, \xi)] = \mathbb{E}_\xi[D(\eta, \xi_t)].$$

Here  $\xi_t$  is the dual Markov process, with finitely many particles in the examples below. Much of the hierarchy of integrable particle systems (ASEP, after Schütz [Sch97] and Borodin–Corwin–Sasamoto [BCS14], with the symmetric case going back to Spitzer [Spi70] (see also Liggett [Lig05, Ch. 8, Thm. 1.1]); the stochastic six-vertex model, after Corwin–Petrov [CP16] and Lin [Lin19];  $q$ -TASEP [BCS14]; the higher-spin vertex models, after Corwin–Petrov [CP16]; and their dynamic deformations) admits dualities traceable to the same quantum group structure. Namely,  $D$  can be built as a matrix element of the universal  $R$ -matrix: a row of vertex weights for an auxiliary line adjoined to the lattice, carried out for the dynamic higher-spin models via the Drinfeld twister of  $U_q(\mathfrak{sl}_2)$  in Kuan–Zhou [KZ23].

Two regimes should be distinguished. *Triangular* dualities build  $D$  from raising operators on the reversible measure, as in [Sch97], Giardinà–Kurchan–Redig–Vafayi [GKRV09], Belitsky–Schütz [BS18], and Kuan [Kua18]; for ASEP and  $q$ -TASEP they collapse a  $k$ -point  $q$ -moment of the height function to a closed evolution of  $k$  dual particles, solved by nested contour integrals (the engine behind the exact one-point laws). They return moments rather than the law, so recovering a distribution still needs a moment problem, ill-posed in the polymer limits. *Orthogonal-polynomial* dualities instead take  $D$  orthogonal with respect to the reversible measure, as in Groenevelt [Gro19], Carinci–Franceschini–Giardinà–Groenevelt–Redig [CFG<sup>+</sup>19], Carinci–Franceschini–Groenevelt [CFG21], Zhou [Zho21], and Franceschini–Kuan–Zhou [FKZ24]; the orthogonality gives direct probabilistic access — Boltzmann–Gibbs principles, after Ayala–Carinci–Redig [ACR18], [ACR21], and correlation identities [CFG21] — without routing through moments.

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